



T.J. Watson Research Center

Integrated Modeling Challenges in *Extreme-Scale Computing*

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Outline of Talk

■ Introduction

- Setting the context: a view of future extreme-scale computing
- What is the primary “wall”: power or reliability?
- Why is pre-silicon modeling a grand challenge in itself?

■ Integrated Modeling

- Power/Temperature, Performance, Reliability
- Levels of Abstraction in Integrated Modeling
 - Relative versus absolute accuracy issues
- Multi-core power and reliability-aware definition; dynamic management
 - Selected examples to illustrate the modeling complexities

■ Concluding Remarks

What is Extreme Scale Computing?

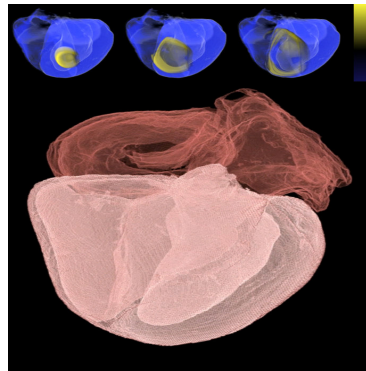
Petascale and Exascale Systems

- Exa- refers to 10^{18} , which is 1000x Peta-
 - Exascale refers to a system that can handle a million trillion operations per second
- Various government agencies have identified exascale as a critical need in the 2018-2020 timeframe
- In *scientific* communities, the important operation is one floating point operation or calculation
 - Exascale in this context refers to 10^{18} flops
 - IBM Roadrunner system: peak of 1 petaflops in 2008
 - Top-ranked system in “Top500” list back in 2008/2009
 - IBM’s Blue Gene product family: L, P, Q systems have consistently been dominant players in the “Top500” and “Green500” lists.
 - So Exascale demands a ~1000x improvement in throughput in 10 years

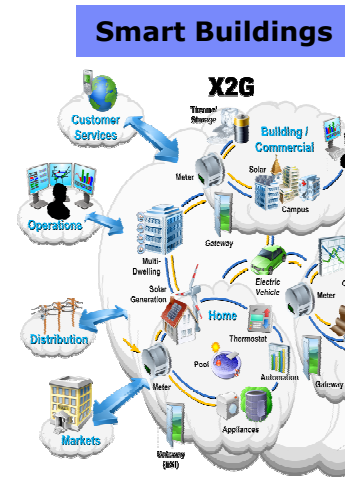
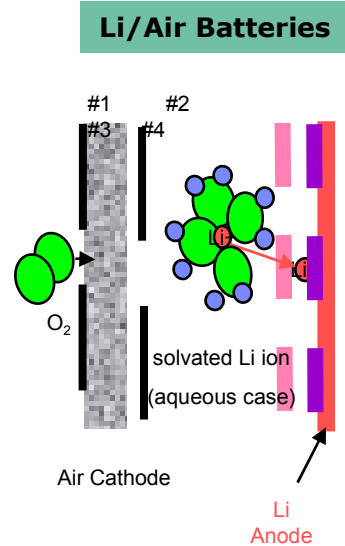
P. Bose, ISPASS-2011 Keynote

Ref: recent tutorial article by Josep Torrellas, “Architectures for Extreme Scale Computing,” *IEEE Computer*, Nov. 2009, pp. 28-35

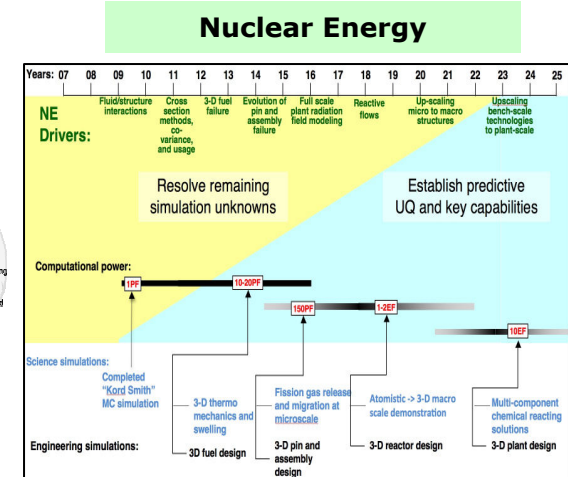
Many Examples of BIG Applications that Need Extreme Scale Computing



Whole Organ Simulation

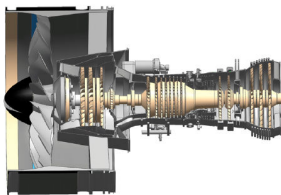


Smart Grid



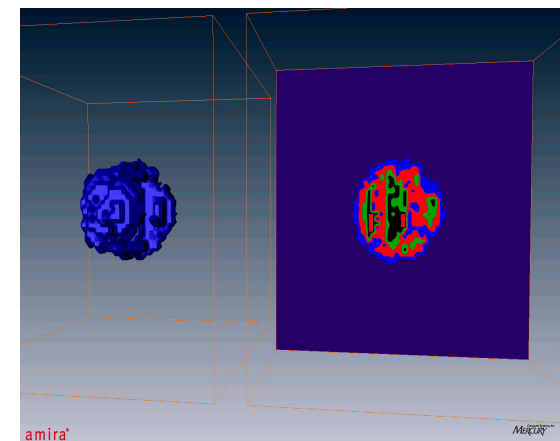
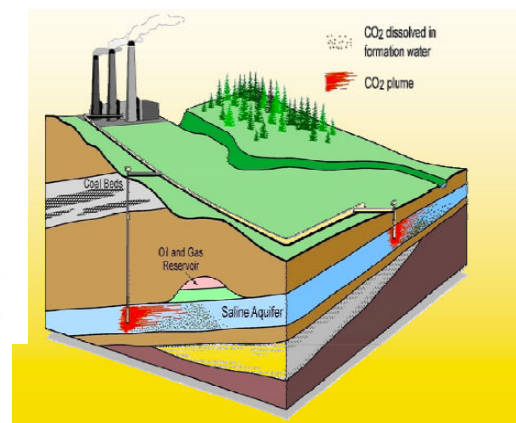
Pratt & Whitney on Intrepid
INCITE PI : Peter Bradley, Pratt & Whitney

- INCITE 2006-2007 technologies now being applied to next generation low emission engines.
- Important simulations can now be done 3X faster
- A key enabler for the depth of understanding meet emissions goals



Low Emission Engine Design

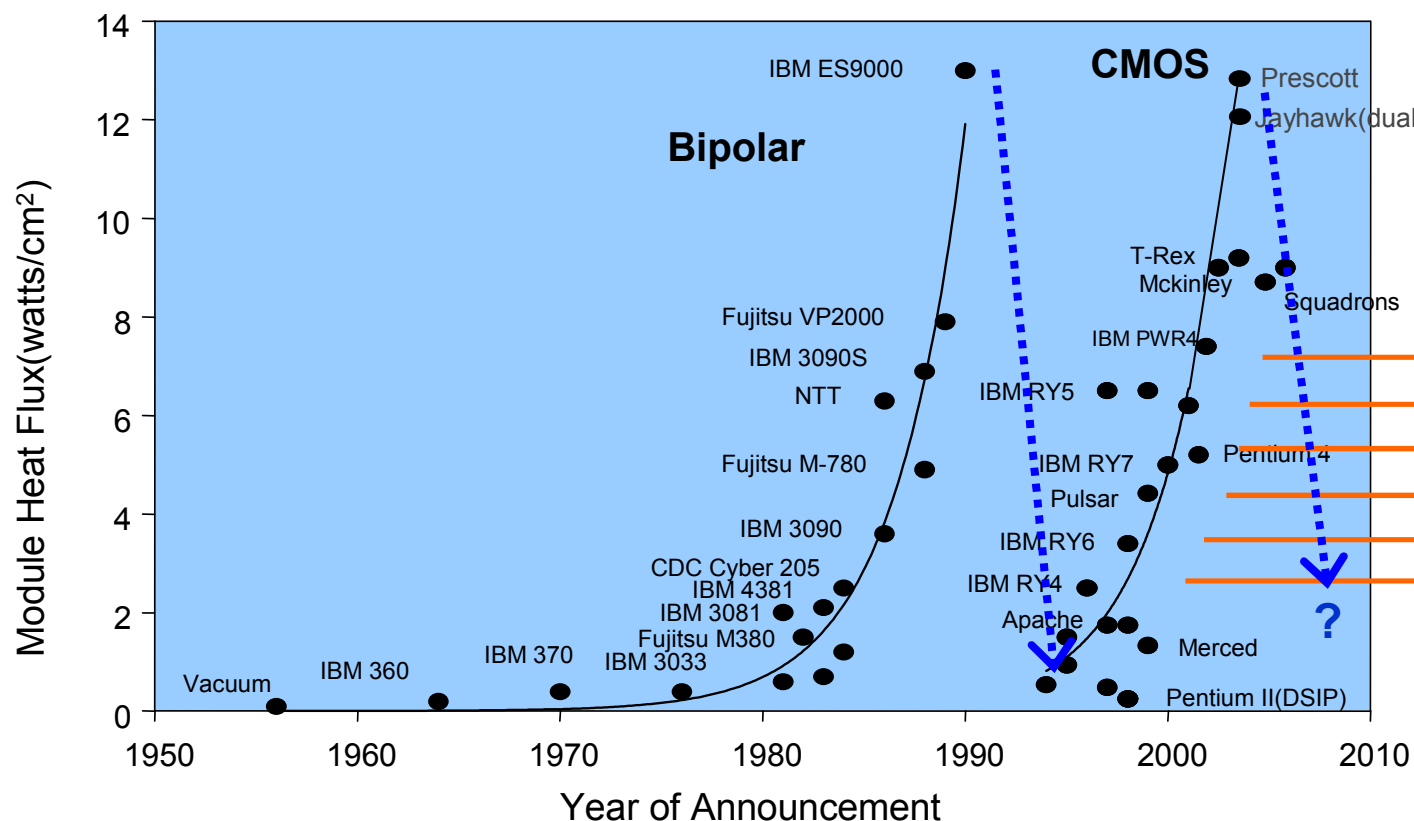
CO2 Sequestration



Tumor Modeling



The Power Wall → Transition to New Technology



Opportunity
for 3D Si

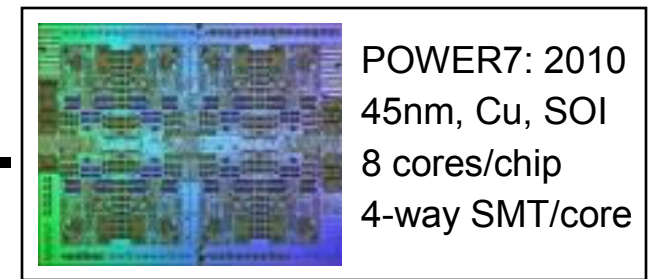
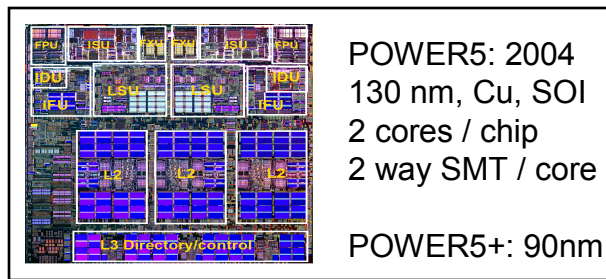
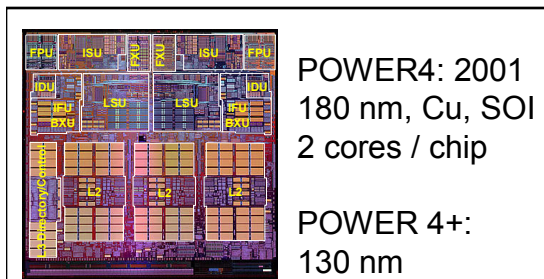
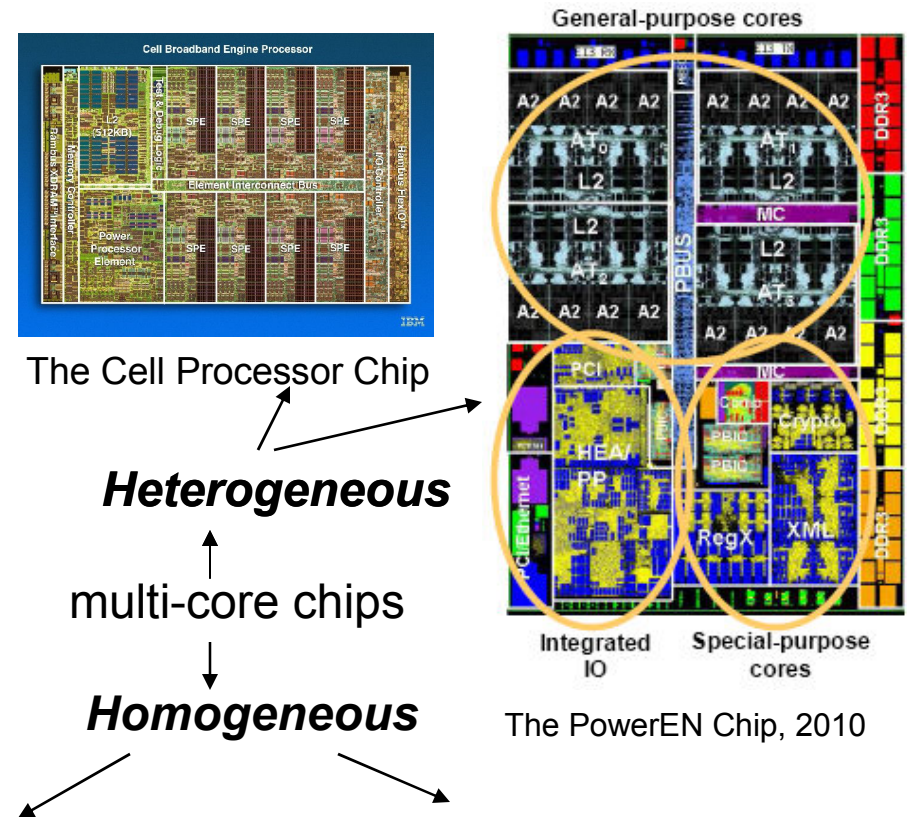
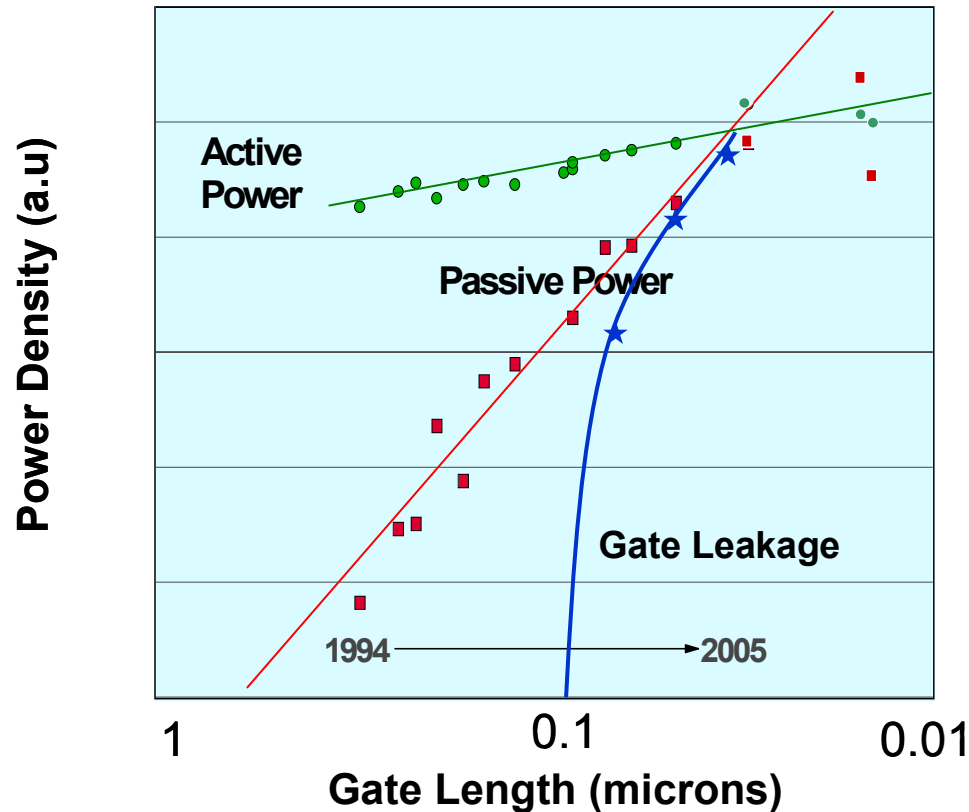
Bipolar to CMOS Transition

Power	15X ↓
Density	50X ↑
Transistor Speed	3-4X ↓

Traditional CMOS to 3D CMOS

Power	10X ↓
Density	3-10X ↑
Transistor Speed	3X ↓

Power-Performance Wall → *Multi-Cores for the Processor Chip*



The Power Wall: A View of the Supercomputer Arena

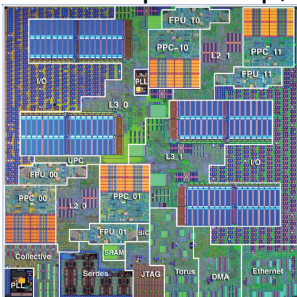
Oxide thickness is near the limit in late CMOS design era

- Density improvements will continue but... power efficiency from technology will only improve very slowly.
- Historic trend of power efficiency improvement will slow

Rank	MFLOPS per Watt	KiloWatts	Supercomputer Location	Brand
1	722.98	59.49	Germany	IBM [QPACE]
1	722.98	59.49	Germany	IBM [QPACE]
1	722.98	59.49	Germany	IBM [QPACE]
4	458.33	276	DOE/NNSA/LANL (USA)	IBM [BladeCenter QS22]
4	458.33	138	IBM Poughkeepsie (USA)	IBM [BladeCenter QS22]
6	444.25	2345.5	DOE/NNSA/LANL (USA)	IBM [BladeCenter QS22]
7	428.91	51.2	Japan	
8	379.24	1484.8	China	
9	378.77	504	United Arab Emirates	IBM [BlueGene/P]
9	378.77	252	France	IBM [Blue Gene/P]

Data from: <http://www.green500.org>

BG/P Compute Chip, 2007



- 4 PPC-440 cores, 850 MHz
- IBM 90nm CMOS ASIC
- 173 sq. mm.
- 208 million transistors
- 16 W

System-on-a-Chip (SoC)



Blue Gene Supercomputers



National Medal of Technology & Innovation
October 2009

Nov 2009 Green 500 List:

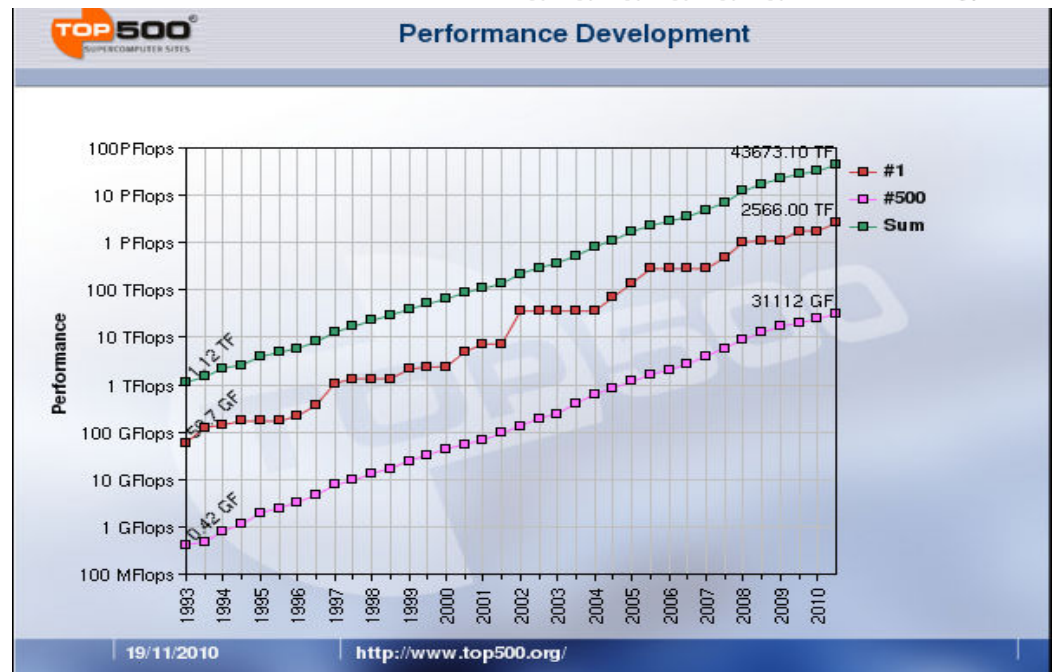
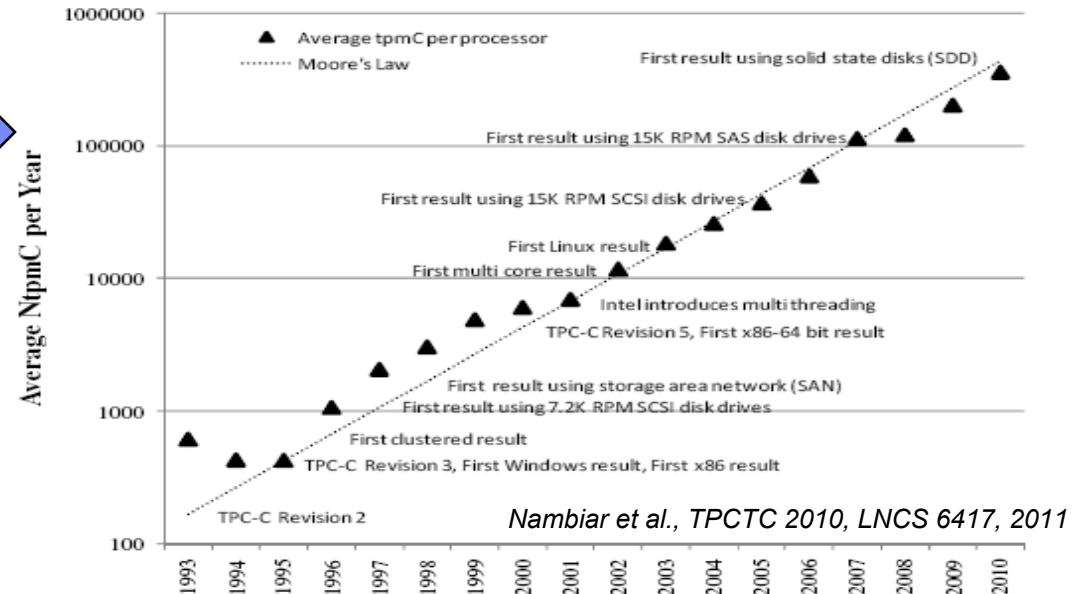
If the world's most power efficient supercomputer is extrapolated to a sustained Exaflop (by 2018), power would be ...

~ 2 GigaWatts

IBM has been a leader in large systems energy efficiency, but meeting the exascale goals is nothing short of a very grand challenge!

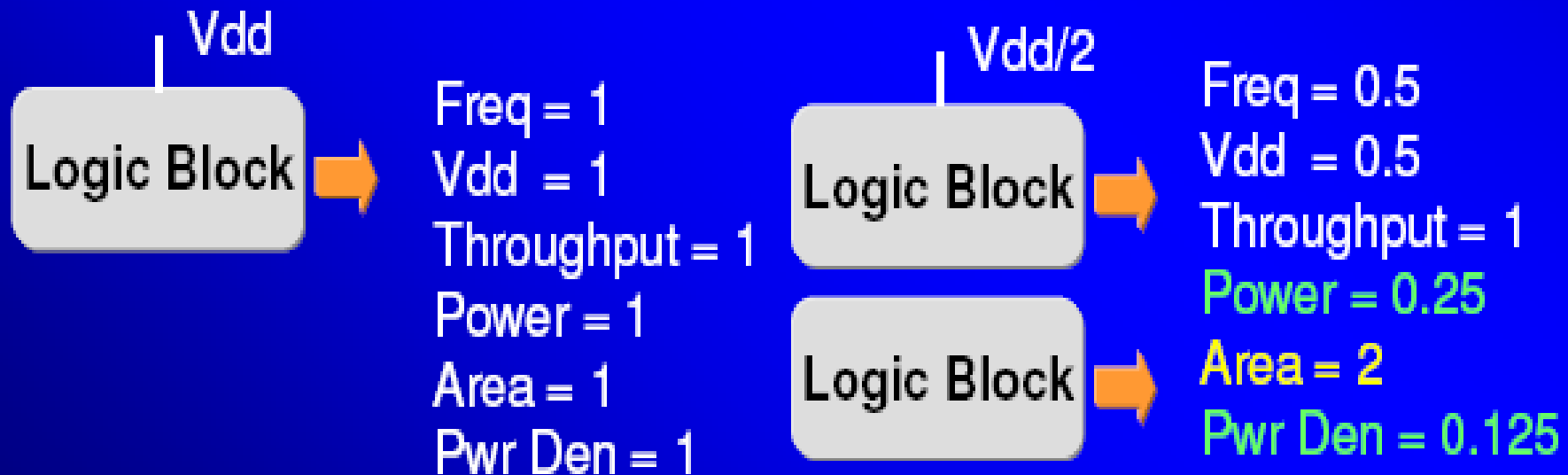
Hybrid Systems – Workload Optimized

- General purpose commercial servers have been on a *2X performance every 2 years* curve
- But **special-purpose** HPC supercomputers have been on a *~4X performance every 2 years* curve
- Power-efficient accelerator sub-cores for **special-purpose** functions constitute the vision of workload-optimized hybrid systems of the future – esp. in emerging new application domains
 - Games market and the Cell multi-core heterogeneous chip was an early trend setter*



Active Power Reduction via Concurrency: The Classic Argument

Throughput Oriented Designs

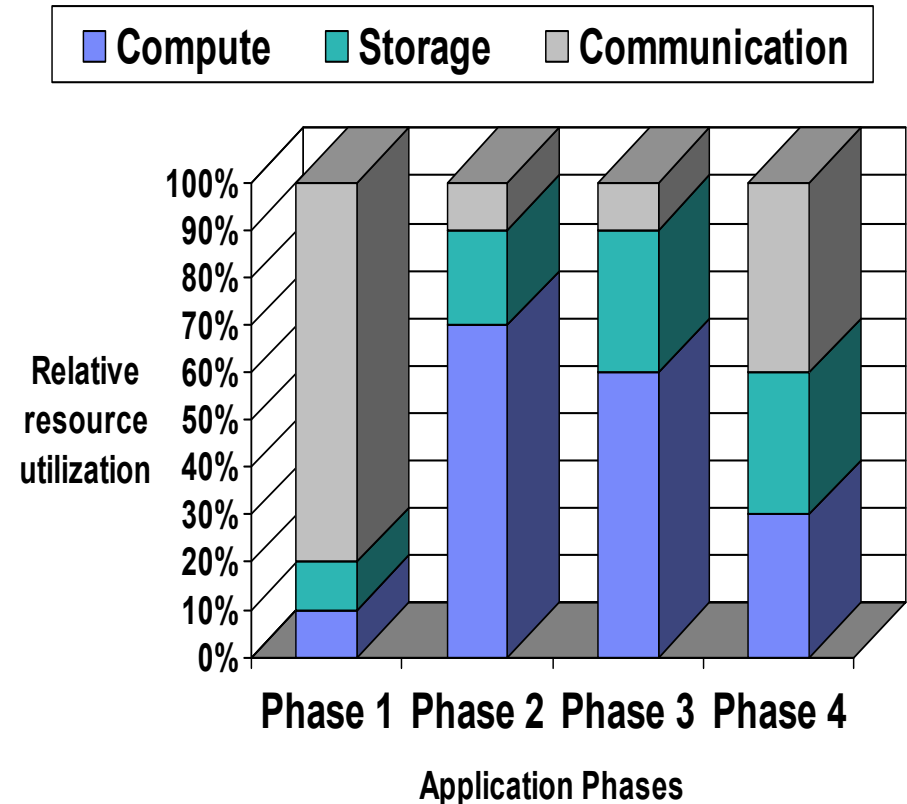
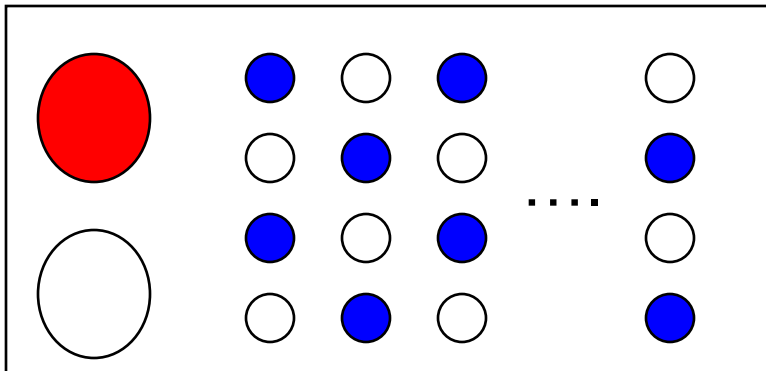


Ack: Shekhar Borkar, Intel, 2005 conf. talk

- A key principle in use in large-scale parallel HPC systems
- Cost constraint for an exascale-regime system implies:
 - manageable number of compute nodes $\rightarrow$ **dozens** of cores/chip
- Also, cannot forget the **serial** (Amdahl) component of HPC codes!

Application-Driven Dynamic Resource Management

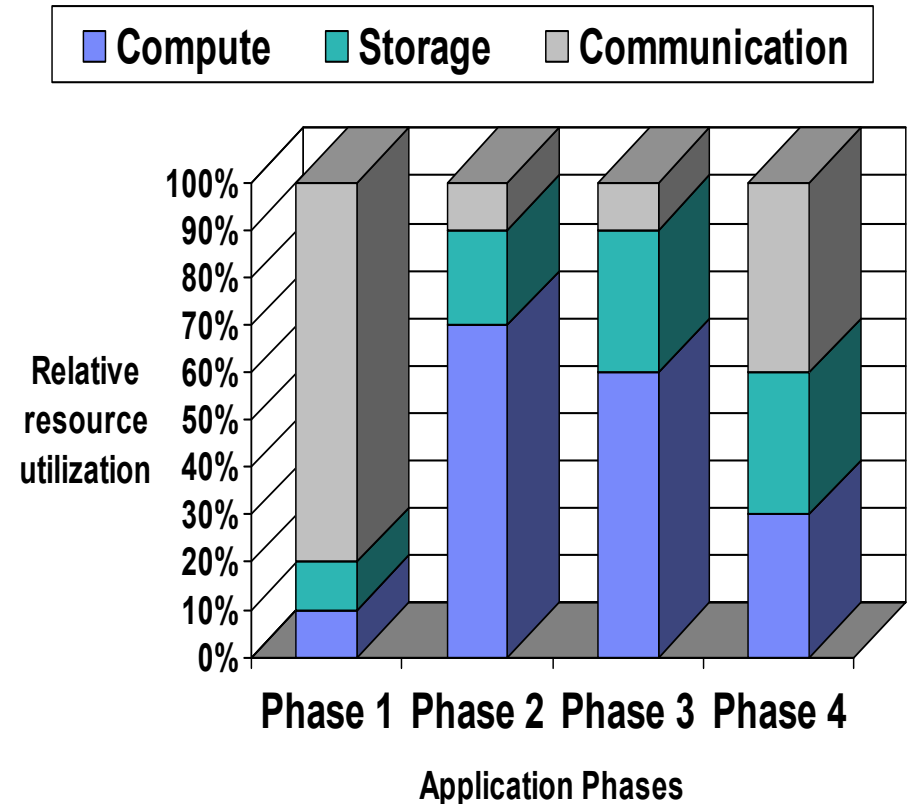
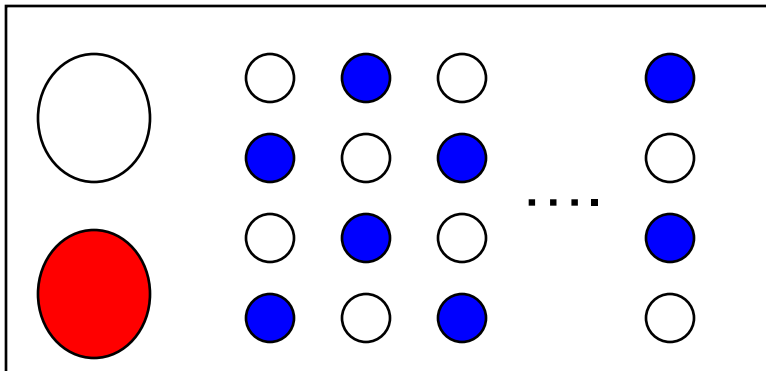
- **Multi-dimensional tradeoff analysis and design space exploration across targeted workloads requires the support of careful, application-driven, dynamic management capability**
 - *Power Shifting* across compute, communication and storage resources
 - Wear-leveling (proactive redundancy) to increase lifetime (MTBF): [J. Shin et al. ISCA-2008](#)



Dynamic power-gating or DVFS features needed to implement power shifting or wear-leveling mechanism

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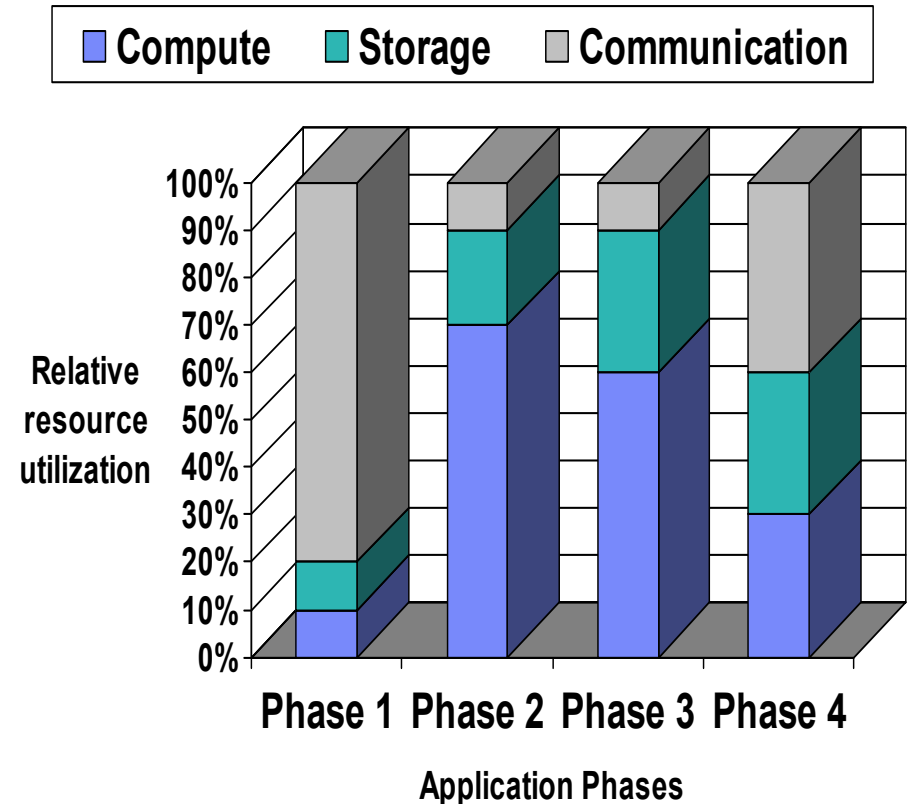
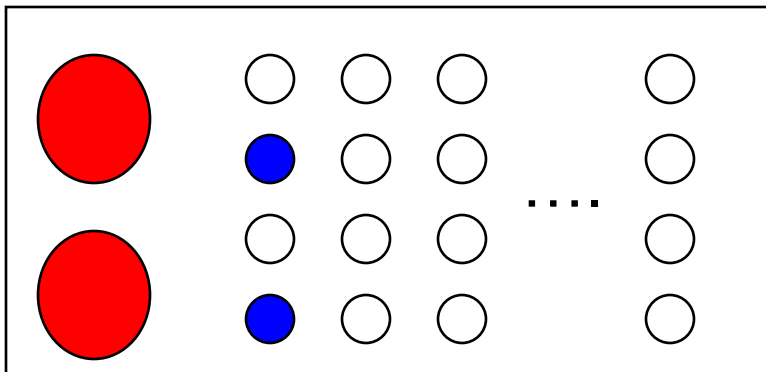


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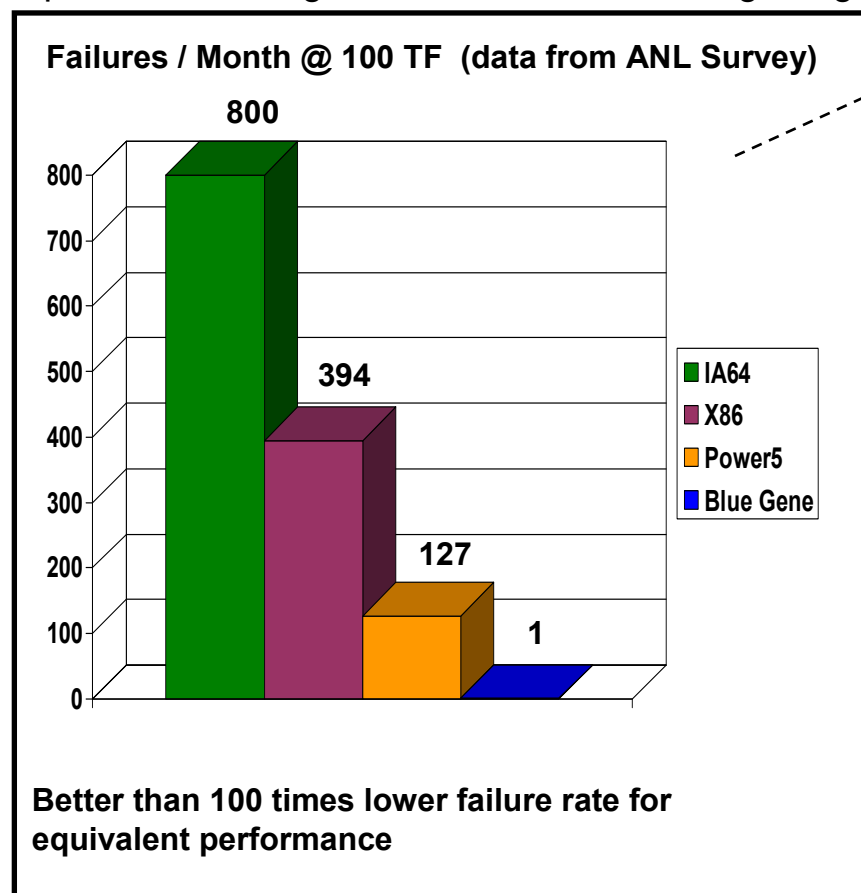
Dynamic power-gating or DVFS features needed to implement power shifting or wear-leveling mechanism

Reliability and Availability: The Other “Wall”



Massive numbers, advanced technologies, and quantity of data produce reliability issues in both hardware and software.

<http://www.er.doe.gov/ASCR/ASCAC/Meetings/Aug06/Stevens.pdf>



Key point: processors targeted for smaller-size systems are usually not suitable for building large-scale supercomputing systems

Hardware Failures

- One million compute nodes, each with a 10 year MTBF would constitute a system that that is likely to fail every 5 minutes

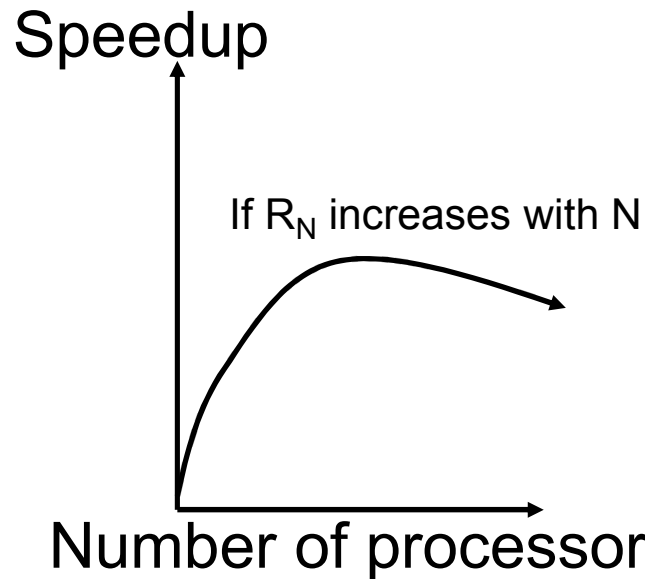
Software Failures

- Brute force techniques (checkpointing) may not be feasible due to disk bandwidth
- Time to checkpoint may dominate computation
- Need to look at reliability at the application level

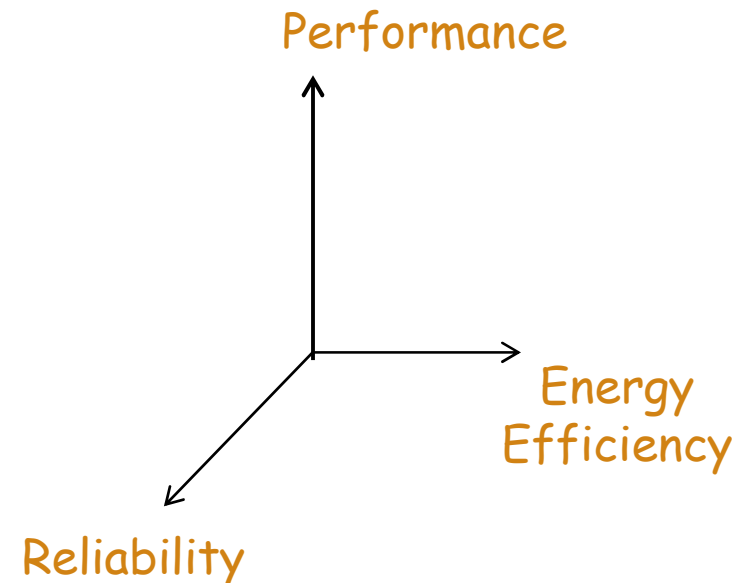
e.g. SWAT project at UIUC (Sarita Adve's group)

*ANL = Argonne National Lab

In fact...reliability is (quite possibly) the *primary* wall !



$R_N = \text{MTTR/MTTF}$ (recovery overhead)
for a N-way system



A reliability-unaware extreme scale design may not even be able to complete a benchmark workload (e.g. Linpack), even with an unconstrained power budget because of too frequent errors and consequent rollbacks!

(See Meeta Gupta et al., MICRO-2009 for local vs. global recovery sensitivities at chip level)

Chip Level Reliability

■ Chip-level functional robustness likely to decline in future

- Increase in transient errors and hard faults
- Maintaining historic levels of chip-level MTBF: cost-prohibitive
- Burn-in difficulty, cost due to high power regime
- Thermal hot spots are a new source of transient/hard failures

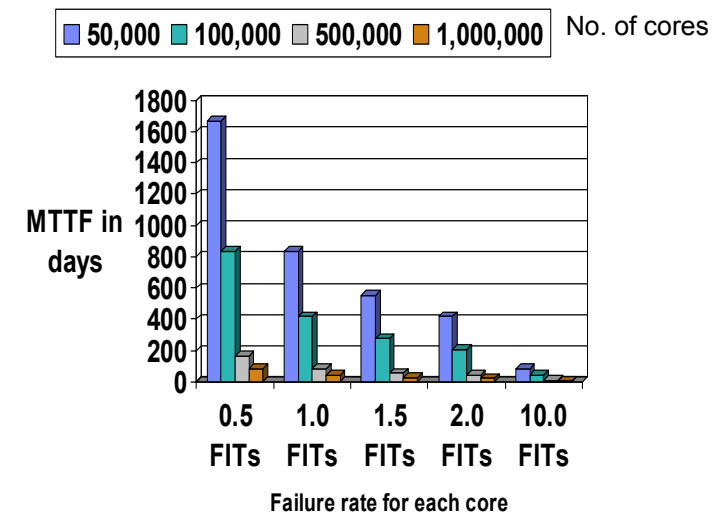
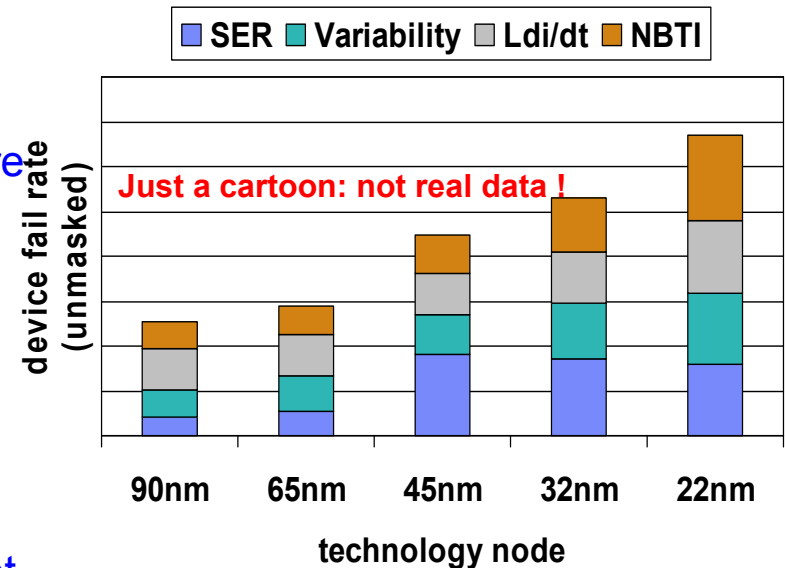
■ System-level reliability targets: going to be hard to meet

■ Two “system” examples:

- SoC with hundreds of core / non-core elements
- Large HPC system with thousands or millions of processor cores/chips [*extreme scale computing*]

■ Need new **cost-effective** solutions across the entire h/w-s/w system design stack to meet FIT targets at any given level of “system” abstraction

- Design and analysis tools must evolve as well



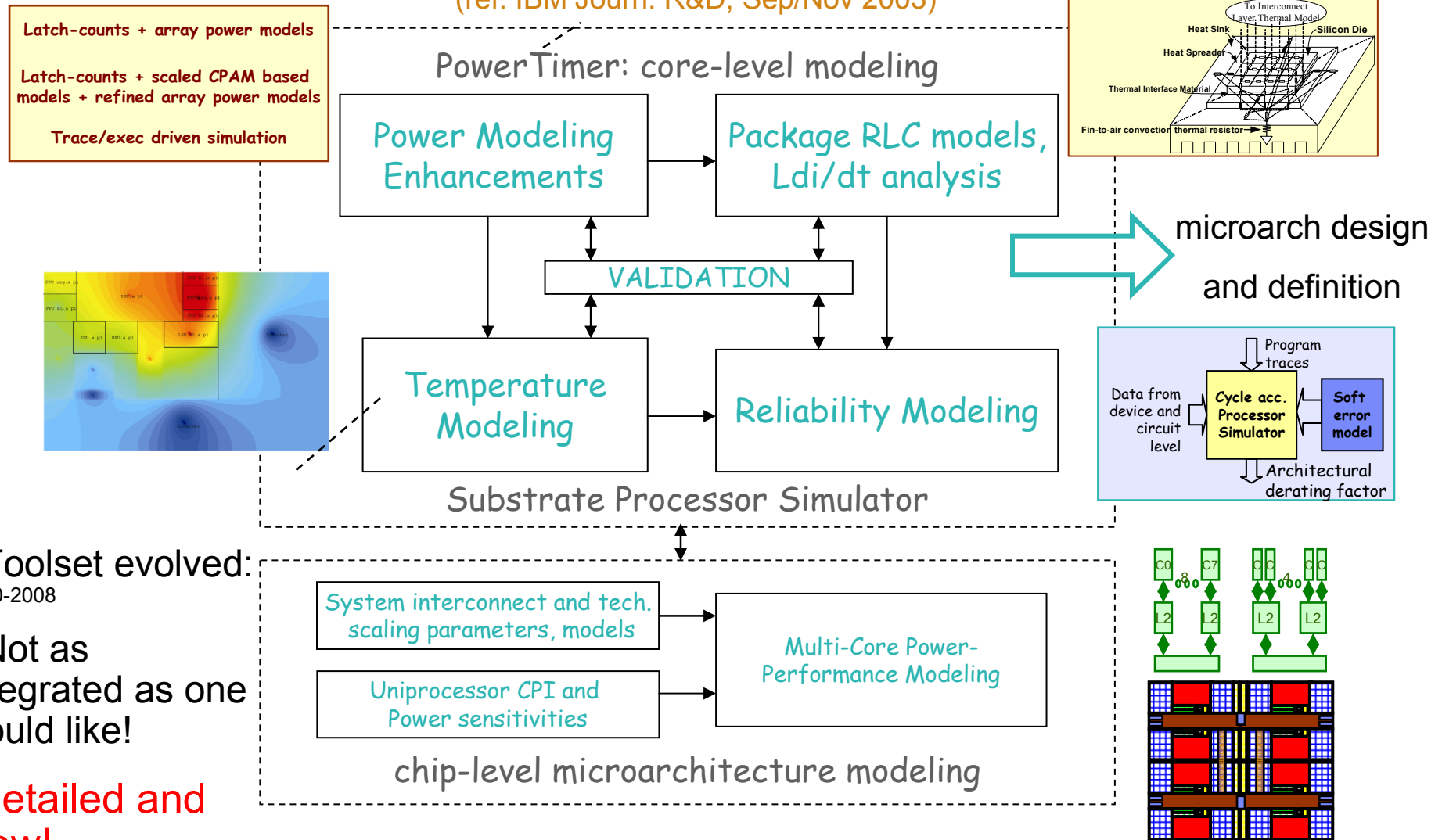
Cost implication trend: not sustainable!

Chip/System Level Definition (Modeling) Approaches

A few specific examples

Towards an *integrated* modeling infrastructure

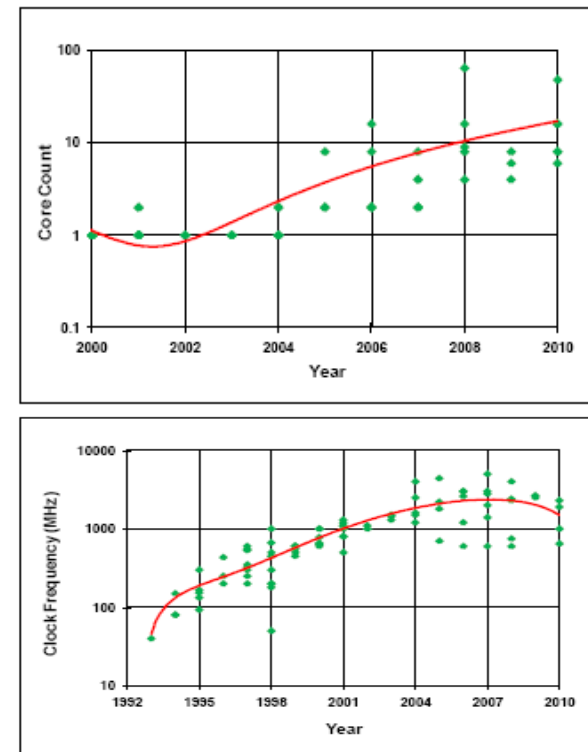
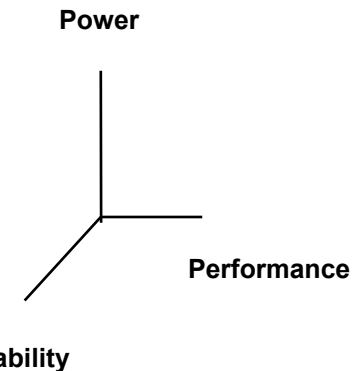
(ref: IBM Journ. R&D, Sep/Nov 2003)



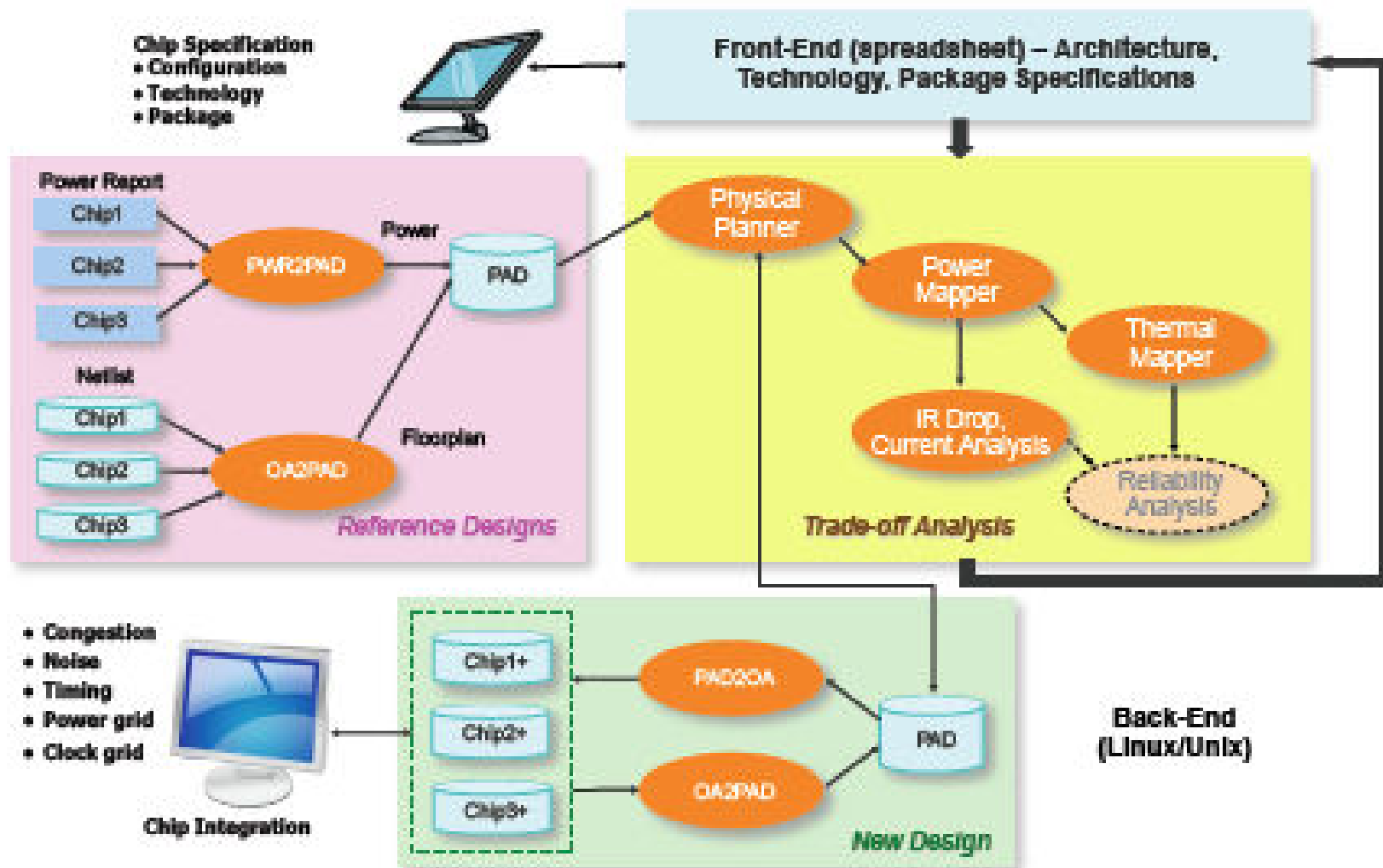
- Toolset evolved: 2000-2008
- Not as integrated as one would like!
- Detailed and slow!

The Pre-Silicon Modeling Challenge in Extreme Scale Systems

- **Why is this a grand challenge in itself?**
- **Because the constraints are multi-dimensional, interdependent and extremely hard to meet at affordable cost. Example:**
 - 20 MW system power
 - 1 exaflops sustained performance
 - MTBF of at least two weeks, preferably 1 month
- **And, because cycle-accurate simulation speed is not scaling up**
 - Host hardware (simulation platform) speed is not increasing
 - Number of cores and target MIPS is increasing exponentially
 - Cycle-accurate performance simulators are very hard to parallelize



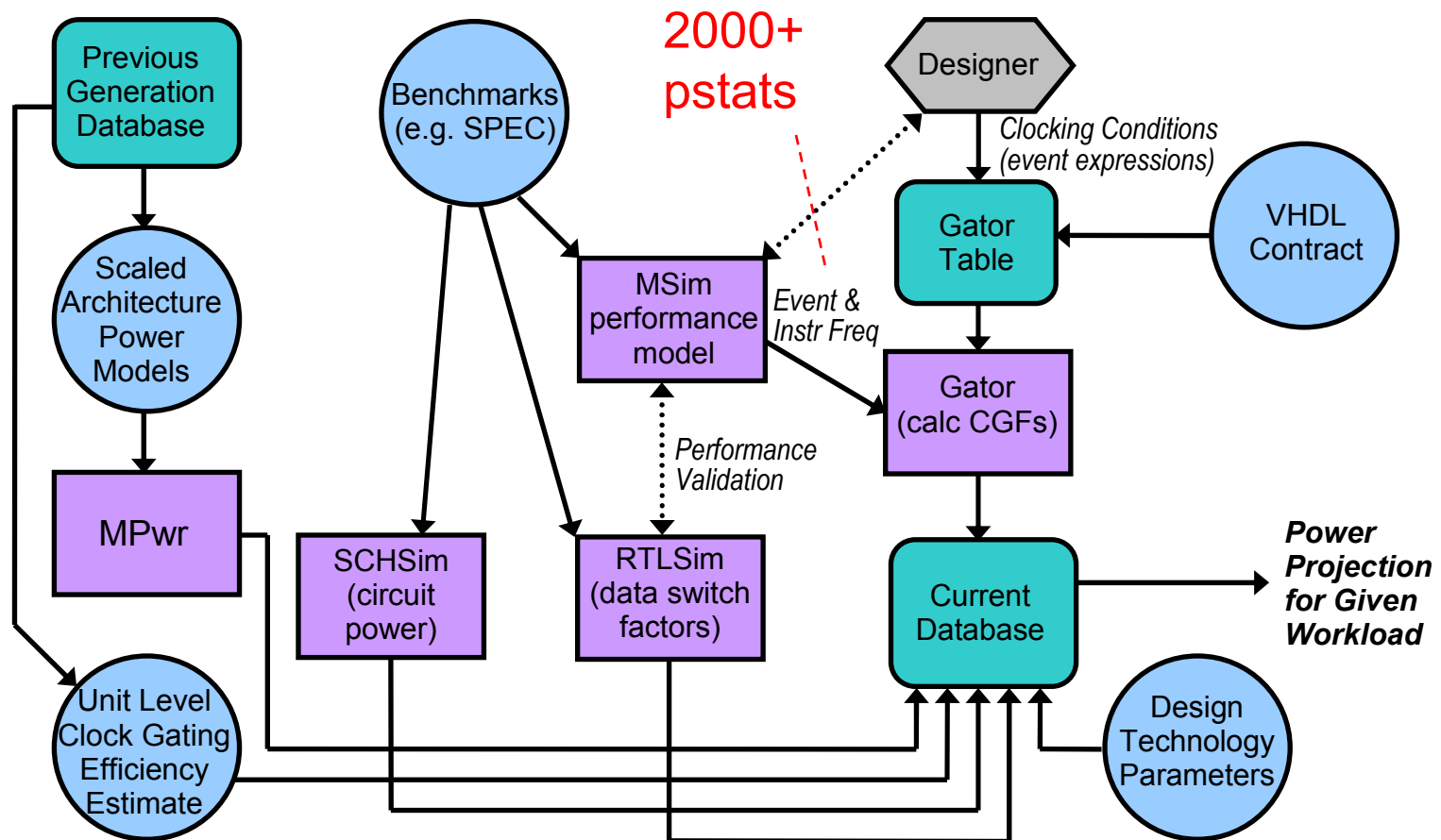
Early Chip Planner Framework at IBM Watson



A step toward better integration of component models

Phased Power Modeling Methodology

■ Concept → HLD → Implementation Phase



H. Jacobson et al., HPCA-17, 2011

Power Model Requirements in the Many-Core System Era

- **Core-level abstraction is a must (for speed)**

- Facilitates multi-core DPM algorithm studies
- Also, fast power-perf tradeoff analyses for core

- **But... detailed reference model useful for macro-wise power budgeting and tracking**

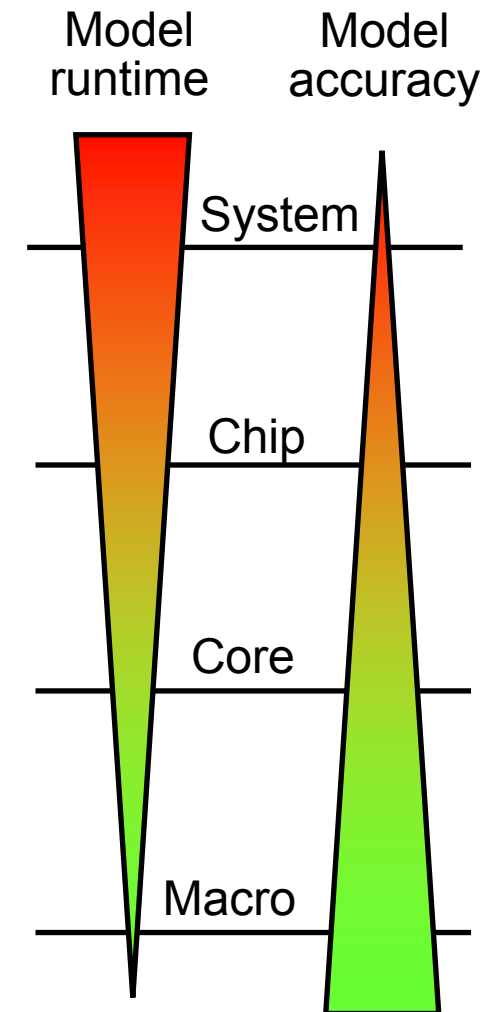
- Core power projection accuracy is important

- **Linear regression based abstraction is a very useful technique**

- [H. Jacobson et al., HPCA-17, 2011](#)
 - See also: previous work: Powell et al. (HPCA 2010), Lee and Brooks (ASPLOS 2006)

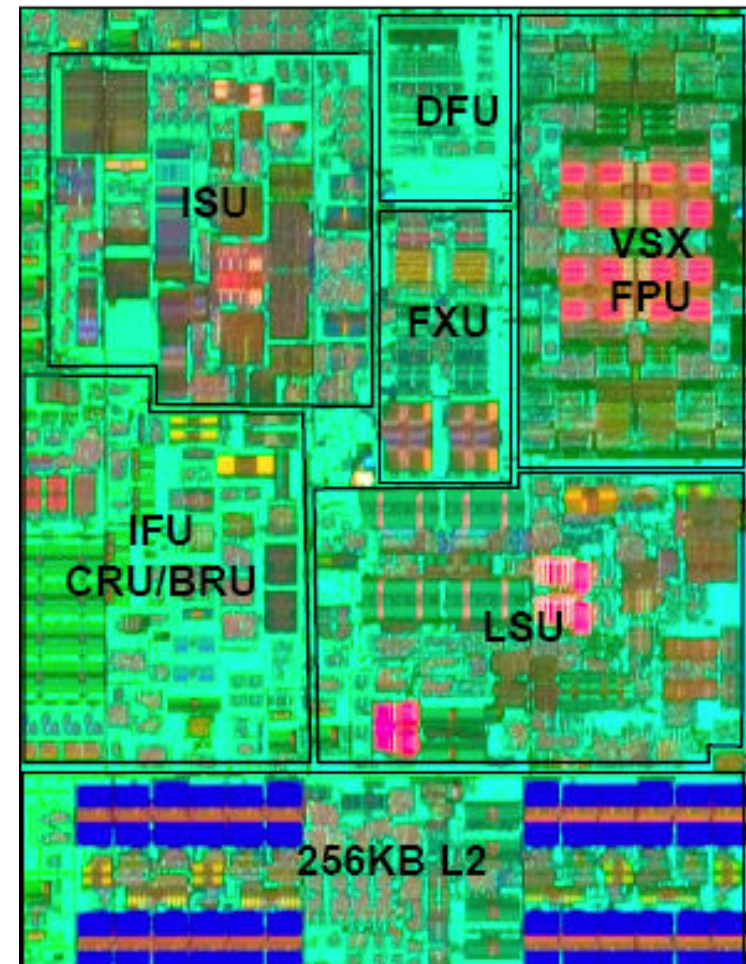
- **POWER7 chip-specific model**

- Detailed p7 reference power model
- Formal attribute selection method
- Support for microarchitecture scalability



Reference Power Model

- **p7 microprocessor chip**
 - High frequency aggressive superscalar out-of-order design
 - 32-thread, 8 core, 32kB I/D caches, 256kB L2 cache
- **p7 core reference power model**
 - Suitable for macro-level power analysis, tracking
 - 2300 μ arch stats
 - 500 RTL macros
 - 2800 modeled clock/port/data gating domains

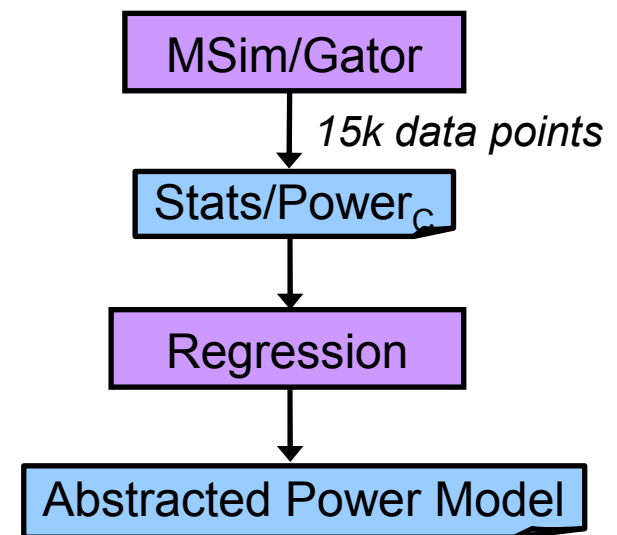


POWER7 (p7) Core + L2

Power Model Abstraction

■ Abstract model obtained through linear regression

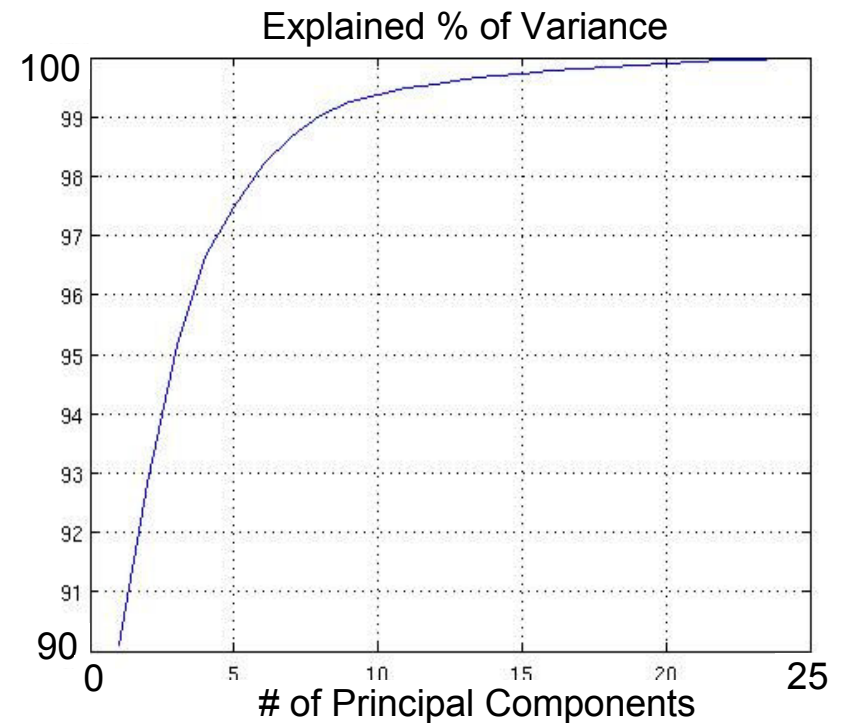
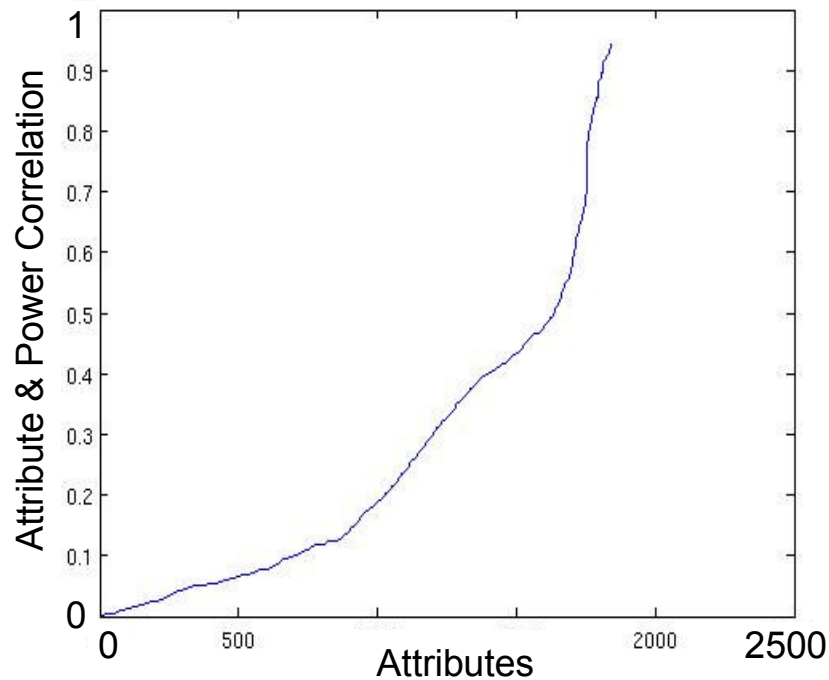
- 15,000+ sets of event stats obtained from simulation of Spec2k6, Commercial, Multimedia, and other workloads
- Power calculated using reference model for each set of event stats
- Linear regression performed to create abstract power model
 - $power = C0 + C1*S1 + \dots + Cn*Sn$
- 10/90 coverage test used to validate the final power model



H. Jacobson et al., HPCA-17, 2011

Attribute Relation to Power Variance

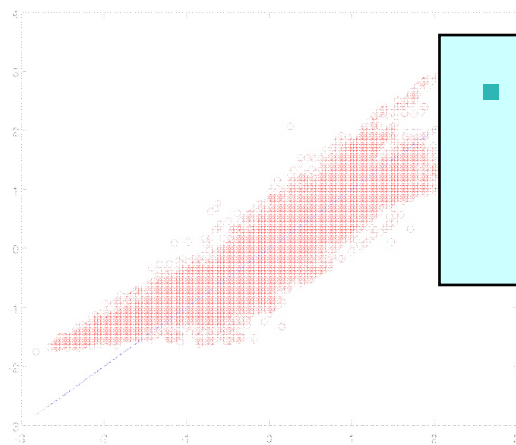
- **A few attributes explain most of power variance**
 - First 8 principal component attributes explain 99% of variance
 - Not necessarily the best for intuitive understanding by humans or ease of implementation



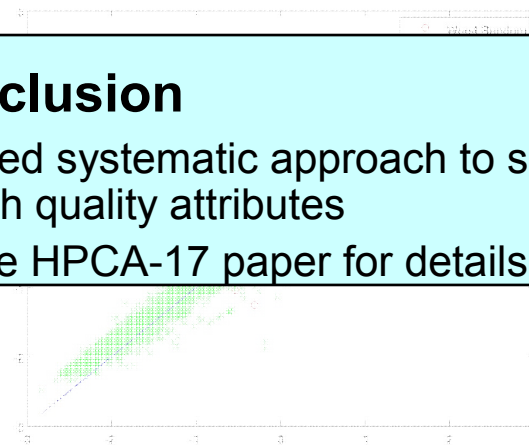
The Importance of Selecting the Right Attributes

- Single attribute (1)
 - IPC fitness corr. 0.905
 - Significant error spread
- Random attributes (6)
 - Best fit corr. 0.976
 - Worst fit corr. 0.109
- Domain experts (6)
 - Expert A fit corr. 0.968
 - Expert B fit corr. 0.971

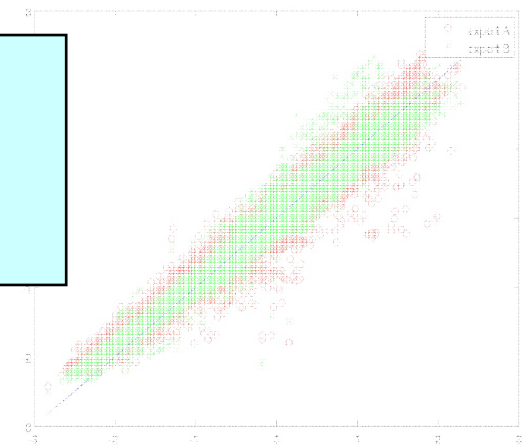
Fitness correlation



Fitness correlation



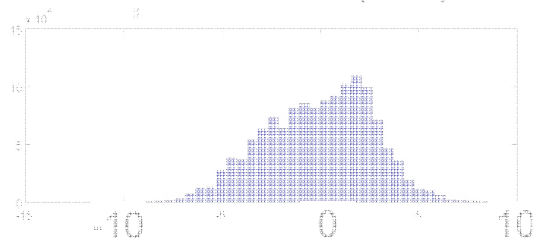
Fitness correlation



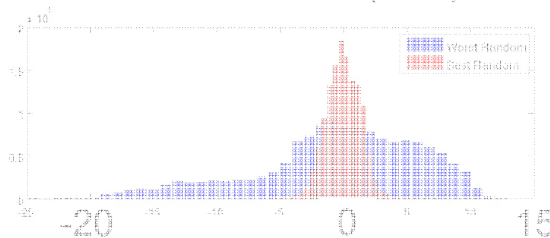
Conclusion

- Need systematic approach to select high quality attributes
- See HPCA-17 paper for details

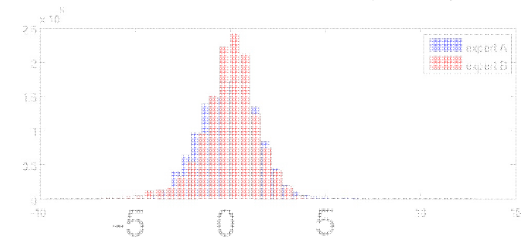
Prediction error (Test)



Prediction error (Test)



Prediction error (Test)

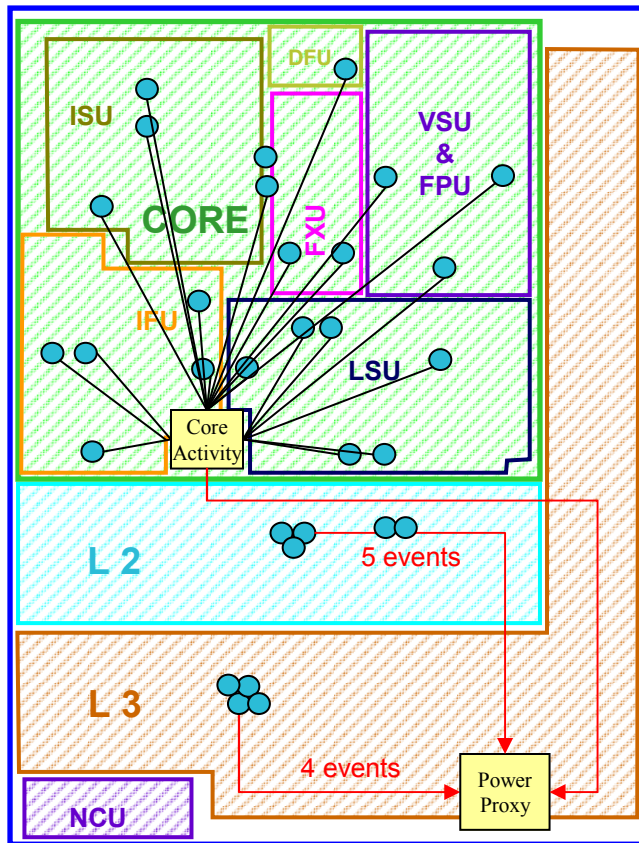


Processor Core Power Proxy: A Hardware Feature in p7

IEEE Micro, 2011 (to appear)

IBM J. R&D, vol. 55, no. 3, 2011

Processor Core Chiplet



● = Activity Sense point

Goal:

Estimate per-core chiplet power that we cannot directly measure

Method:

- For each functional unit, pick small subset of activities to infer power consumption (e.g. cache & regfile reads & writes, execution pipeline issue)
- Weight each activity to represent how much relative power it consumes
- Combine weighted Core, L2, and L3 activity, then add constant offset plus clock grid power to form:

$$\text{Chiplet Active Power} = \sum (W_i * A_i) + C + K*f$$

Result:

- EnergyScale Firmware adjusts this value for effects of leakage, temperature, and voltage

Hardware design was driven by power model abstraction research at IBM Watson (A. Buyuktosunoglu et al.)

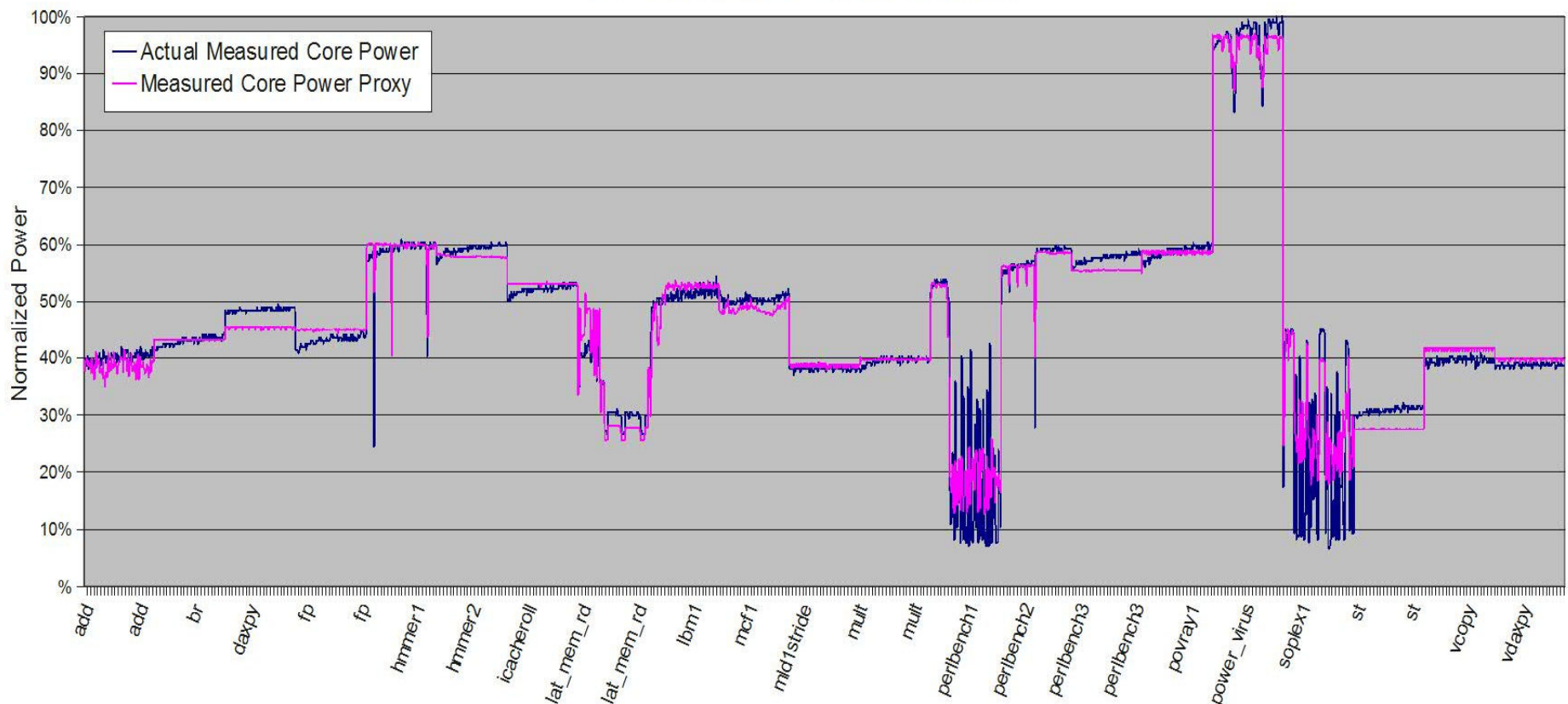
Power Proxy Measurements

IEEE Micro, 2011 (to appear)

IBM J. R&D, vol. 55, no. 3, 2011

- EnergyScale firmware budgets power across multiple processors and memory, used to:
 - Shift power to cores or other components (e.g. memory) that need it the most (Especially important to achieve higher overall performance under a power cap)
 - Enable Server Partition power accounting

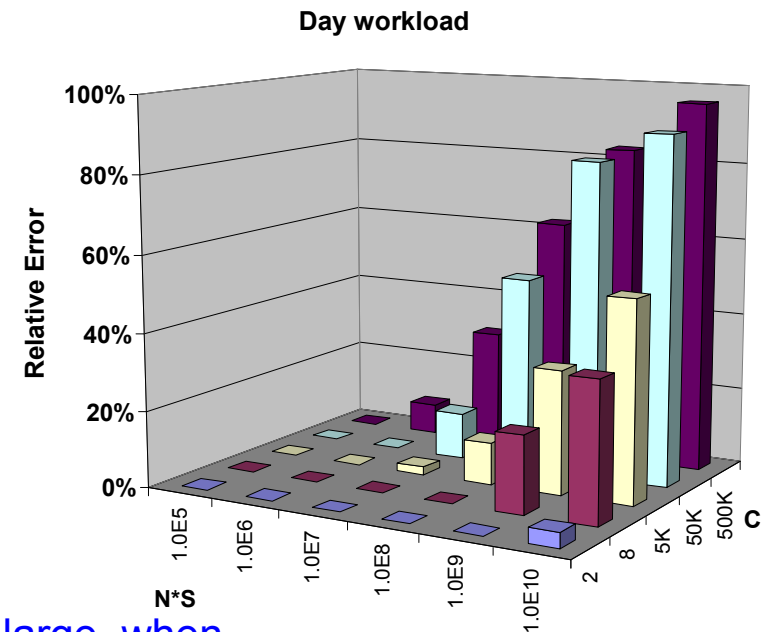
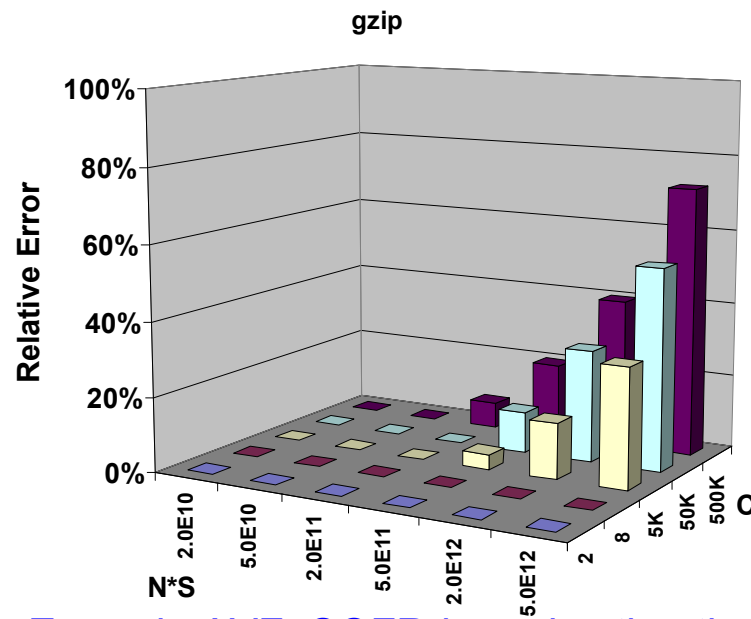
Processor Power Measurements by Workload



Pitfalls of Architectural Abstractions: *An Example from Soft Error Rate (SER) Analysis*

Architectural Vulnerability Factor

$$\text{System SER} = \sum [\text{AVF}(i) * \text{Raw_SER}(i)] \dots \text{AVF + SOFR abstraction}$$

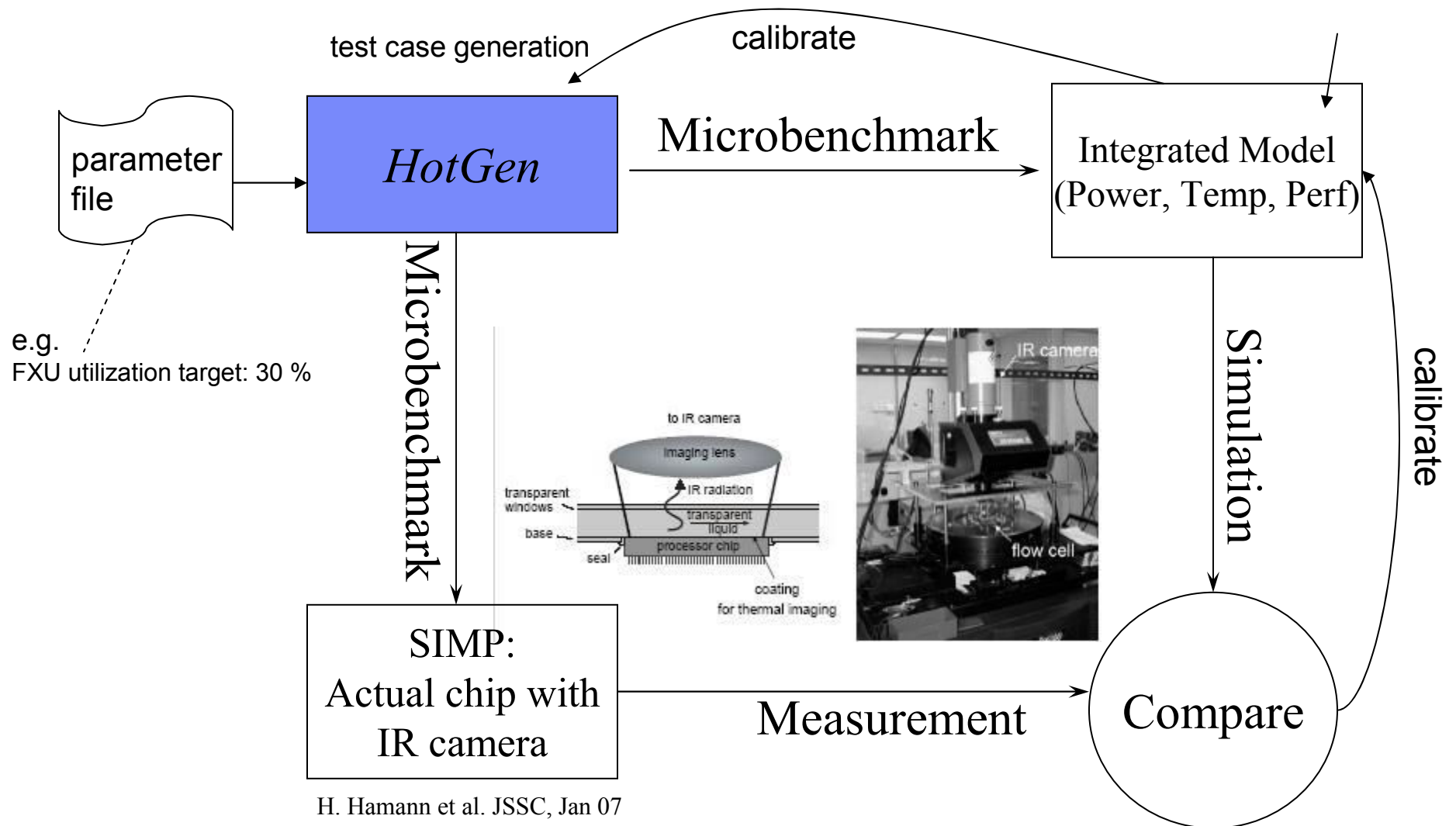


Errors in AVF+SOFR-based estimation get very large, when number of modeled cores, C in the system becomes very large, or if the raw error rate of each of the N cores becomes very large

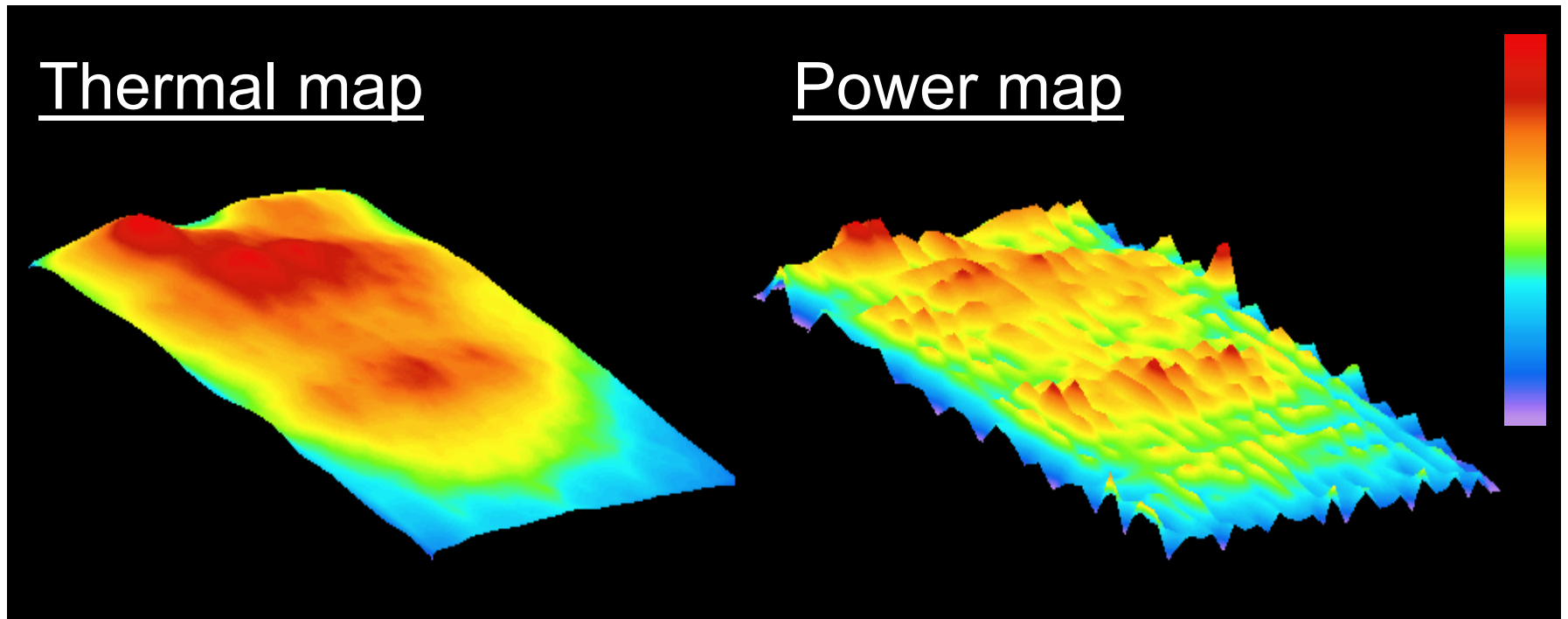
Architecture-Level Soft Error Analysis: Examining the Limits of Common Assumptions,

X. Li, S. V. Adve, P. Bose, and J. A. Rivers, *Proc. of the Int'l. Conf. on Dependable Systems and Networks (DSN)*, June 2007.

Power Model Calibration/Validation Methodology



POWER5 Hotspot Patterns

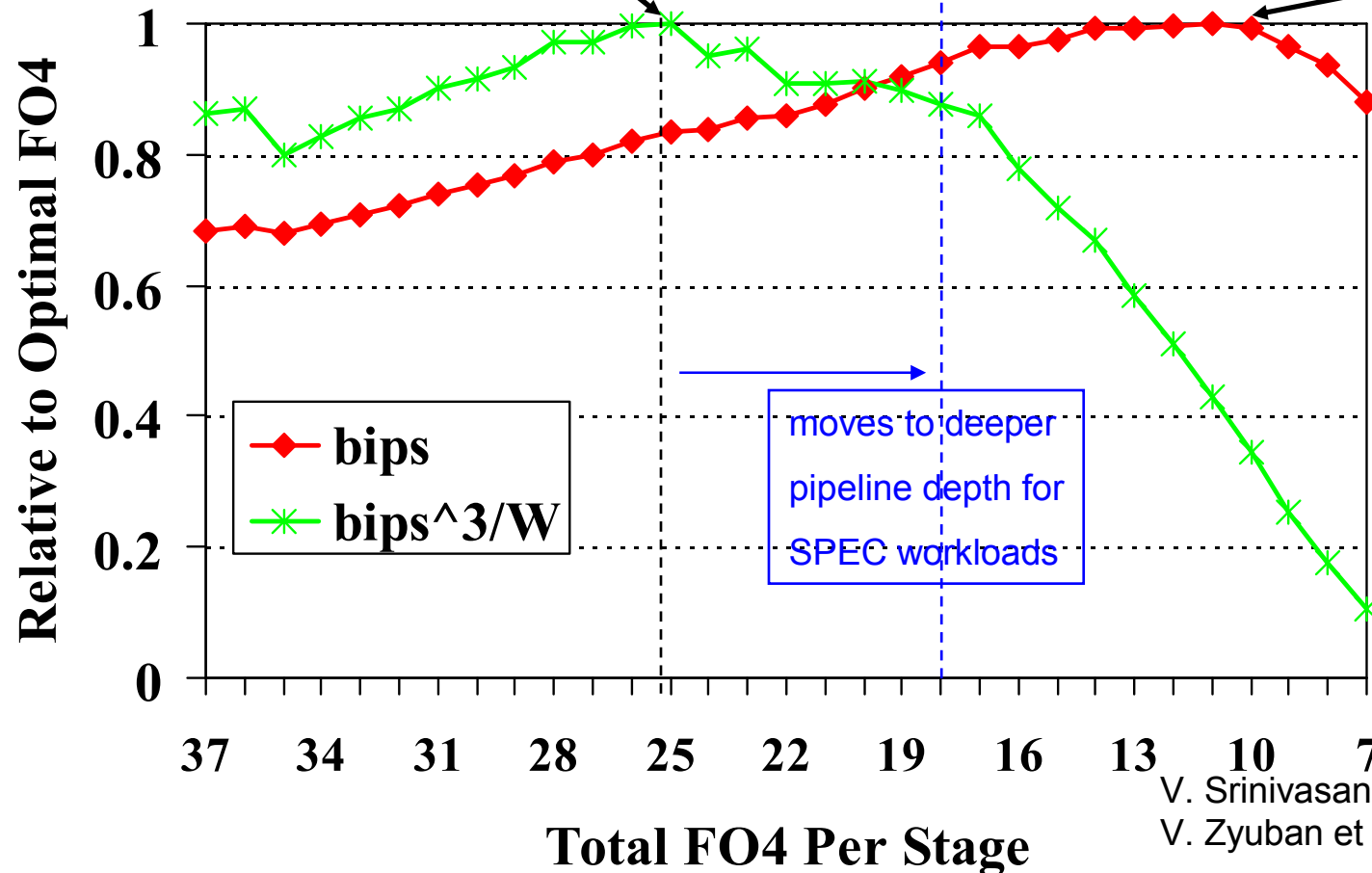


- 50 different workloads for POWER5 imaged & analyzed
 - HotGen microbenchmark generator tool
- observed significant differences in circuit utilization

(H. Hamann et al., ISSCC-2006)

Optimal Pipeline Depth: TPCC Workload

Power-performance optimal Performance optimal



V. Srinivasan et al., MICRO-2002

V. Zyuban et al., IEEE TC, 8/2004

Note: *Optimal point on x-axis is the important output of*

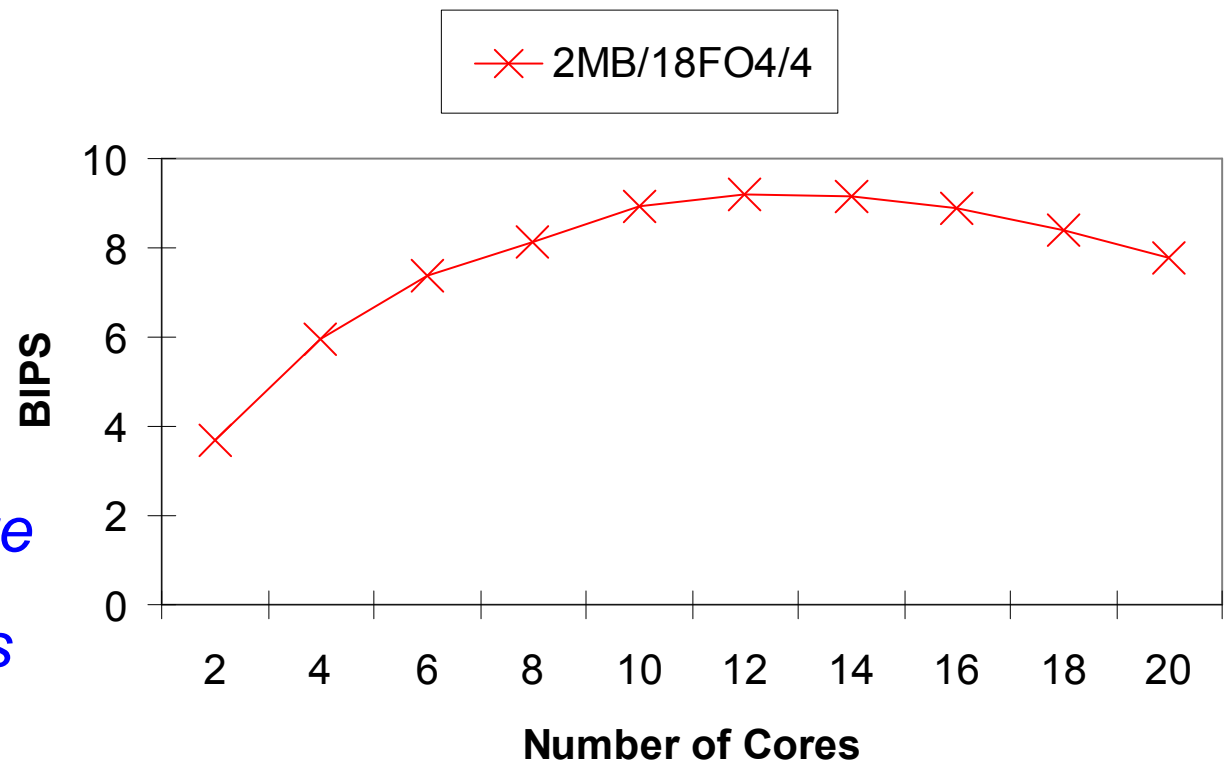
such an analysis model; y-axis value absolute accuracy not very important!

CMP Space Exploration Results

400mm², Cheap Thermal Package, CPU bound benchmark

The optimal core-count for a given core type

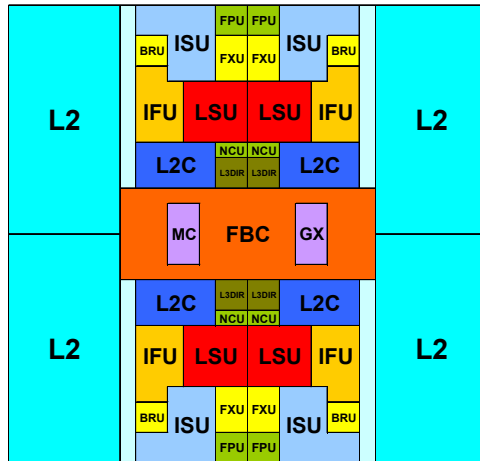
Analytical or hybrid models do quite well in such scenarios



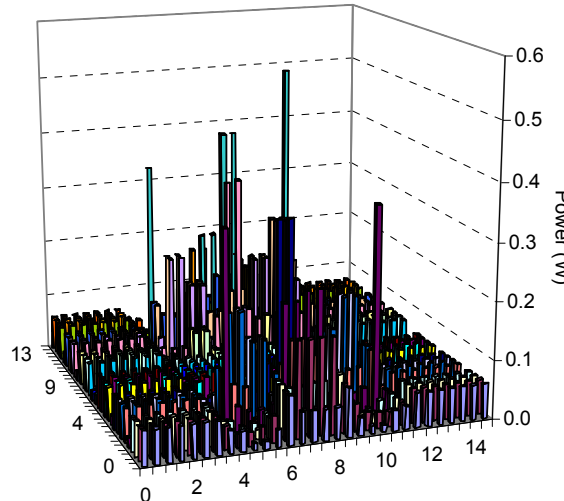
Yingmin Li, Zhigang Hu et al., HPCA 2006

Chip-level Lifetime Reliability Analysis

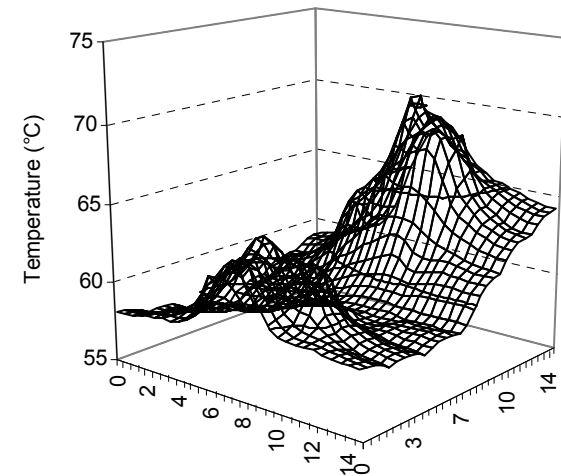
◆ Floorplan



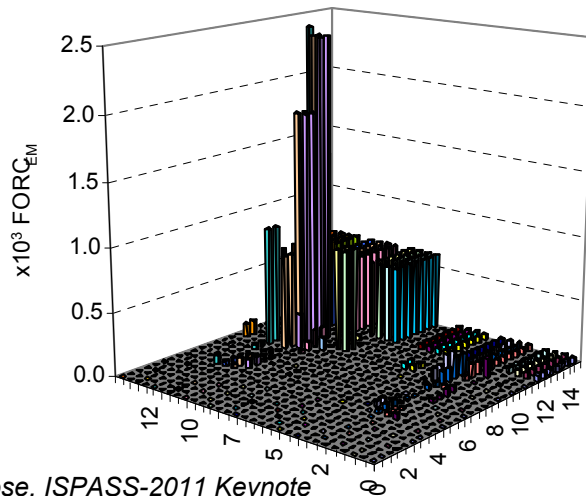
◆ Power



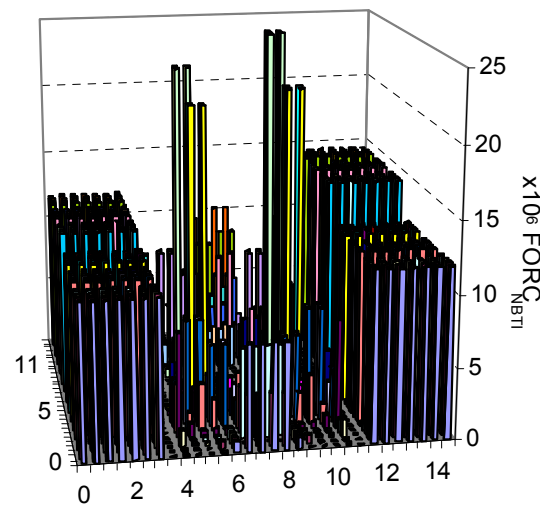
◆ Temperature



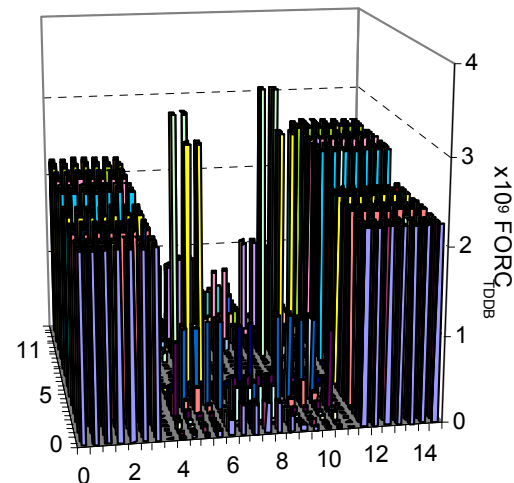
◆ FIT due to EM



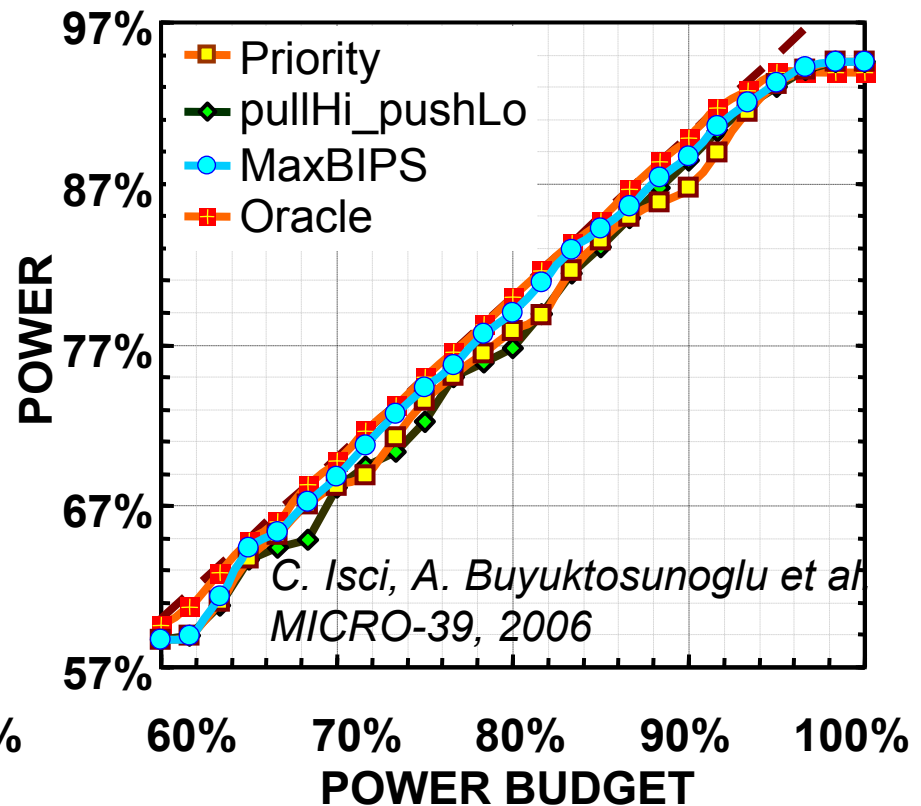
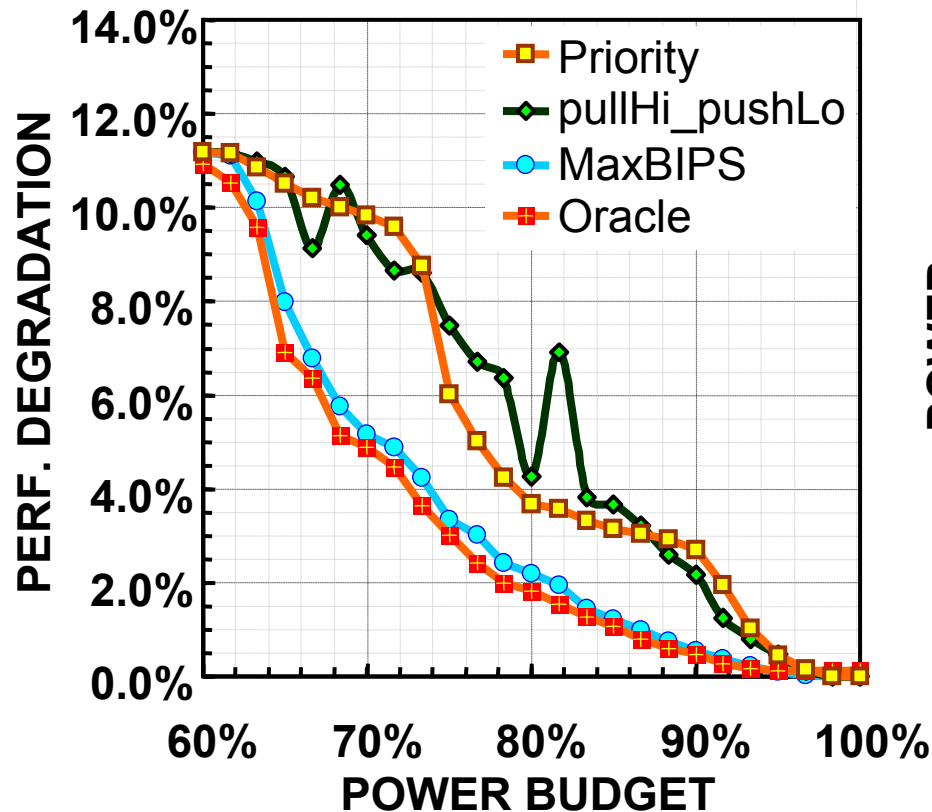
◆ FIT due to NBTI



◆ FIT due to TDDB



Power-Performance Tradeoffs (on-chip, global power management; DVFS): A Key Modeling Challenge!

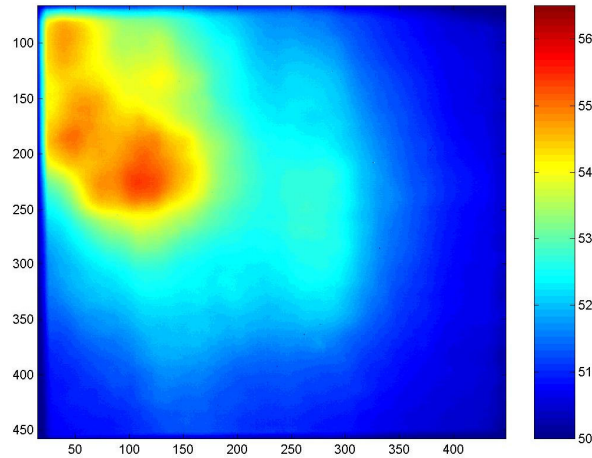


C. Isci et al., MICRO-2006

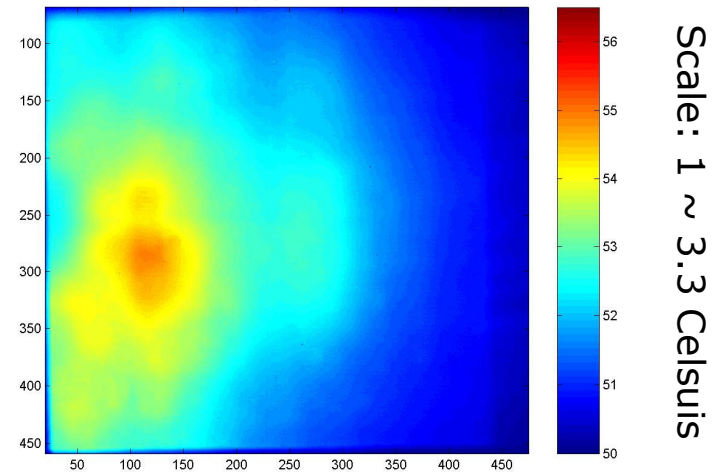
- MaxBIPS within 1% of Oracle
- Verification complexity of multi-core power management algorithms – scalability – is a key issue [A. Lungu et al. MEMOCODE 2009]

Activity migration [*temperature-aware task scheduling*] reduces maximum on-chip temperatures

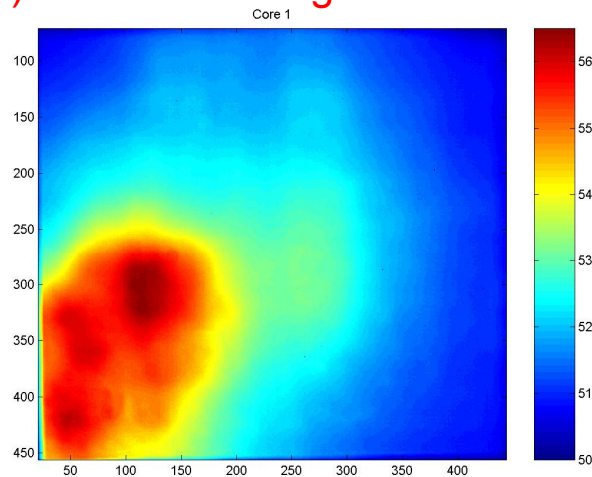
(a) DAXPY running on core 0



(c) DAXPY hopping every 7ms



(b) DAXPY running on core 1

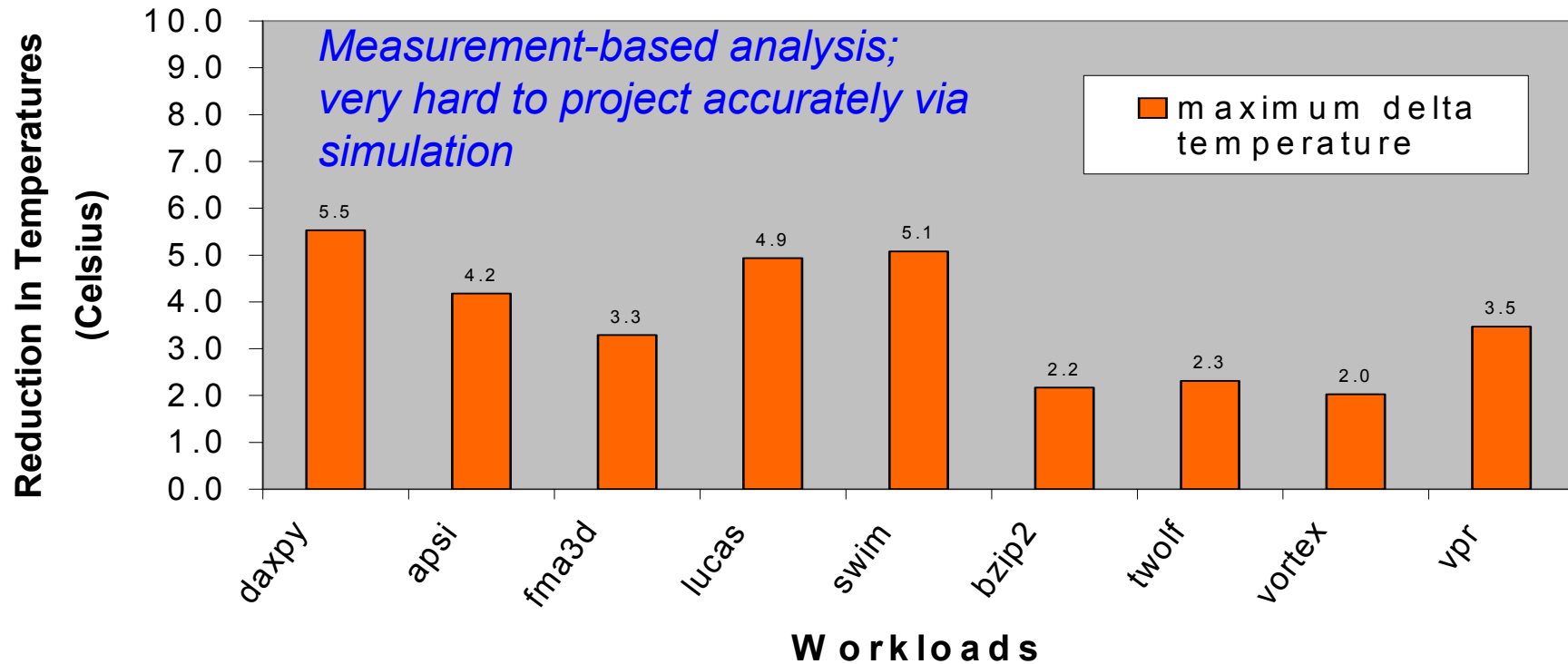


Chip designs could leverage the lower temperatures for higher frequencies, lower-cost packaging or enhanced reliability.

Leveraging Spatial Heat Slack

Activity Migration reduces Hotspots

Summary: Core-hopping (4ms) reduces maximum on-chip temperature

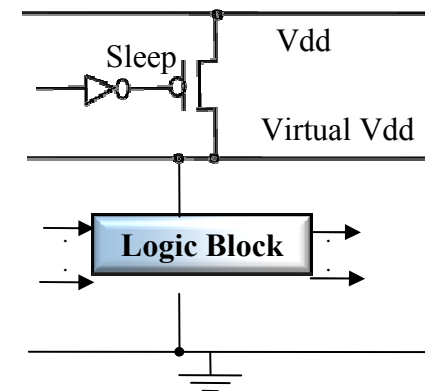


% slow down	0.1	-1.1	-0.5	0.4	1.0	1.1	1.6	0.9	2.5
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Power Gating as a Dynamic Management Knob

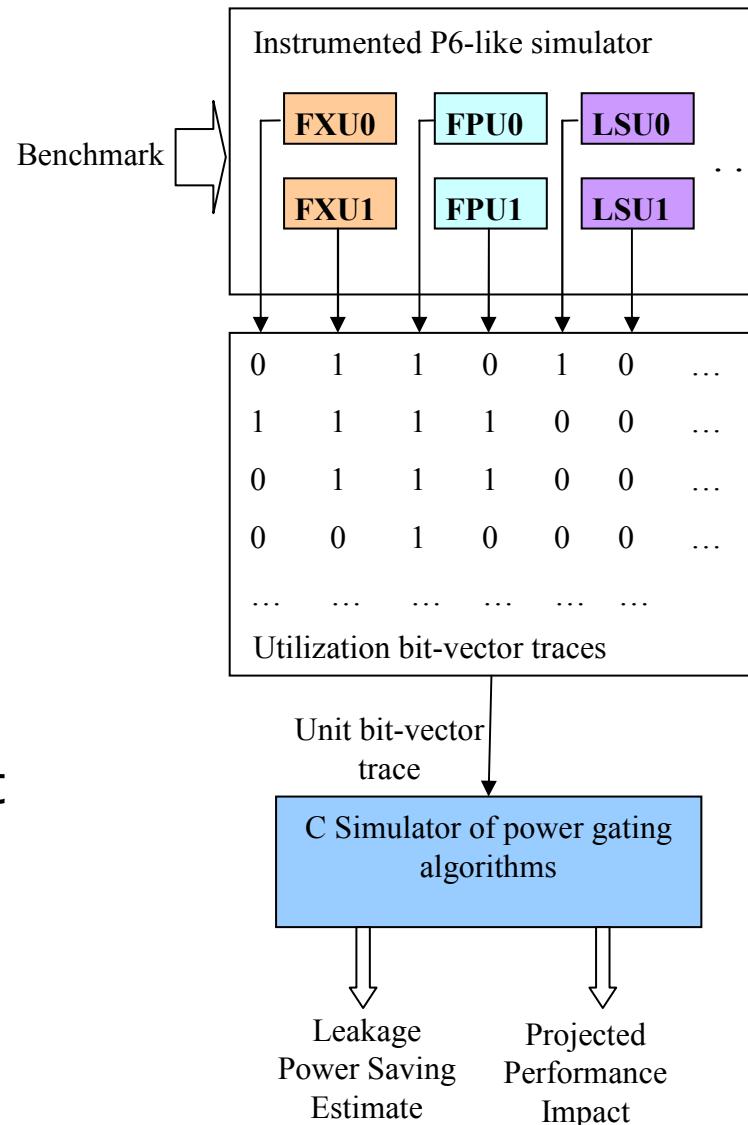
- Power Gating (PG) is becoming an essential actuation knob for dynamic power management
 - ▣ Header or footer transistor gates off power to the “macro” during idle durations
 - ▣ Applied at core-level (per-core PG) or within a core at the unit-level
 - ▣ PG is applicable to a broad range of compute nodes that exhibit variable idle times
 - Mobile, Desktop, Enterprise etc.
- Our end target is efficiency at all levels: from chips, all the way through to the data center level

Header Transistor Implementation



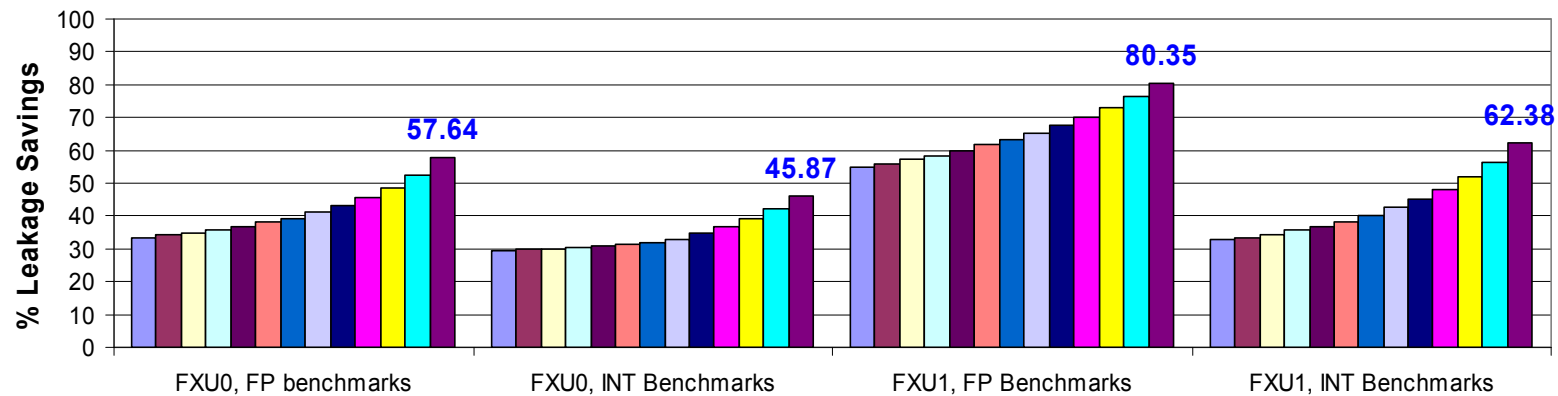
Methodology for Core-Level Power Gating Analysis

- Use bit-vector traces (utilization) from instrumented cycle-accurate perf. simulator
- Workloads: SPEC, other traces
- Implement trace driven simulator for power gating algorithms, obtain:
 - Leakage power savings estimate
 - Projected performance impact
 - Assume constant performance impact of 3 cycles on wake-up

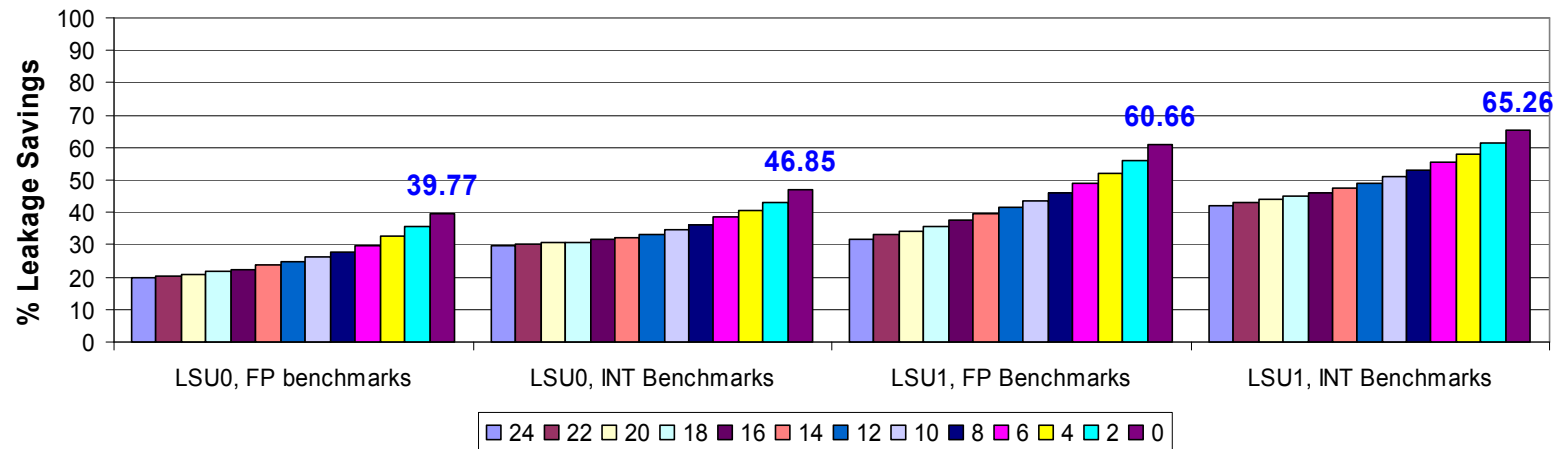


Power Savings Potential for Power Gating of Functional Units

Power gate potential function of break-even point for **FXU0** and **FXU1** units



Power gate potential function of break-even point for **LSU0** and **LSU1** units



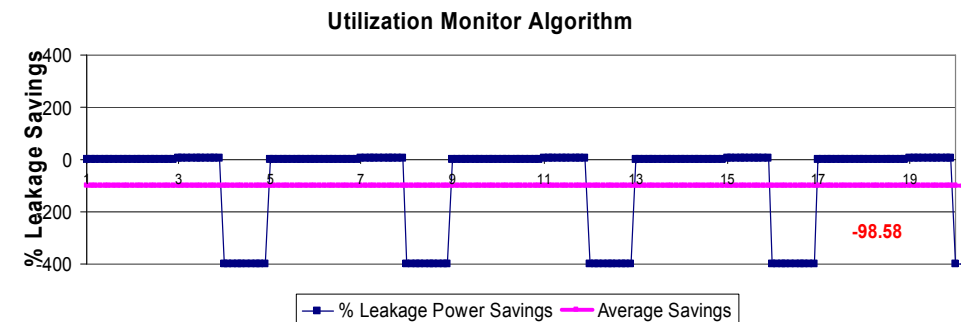
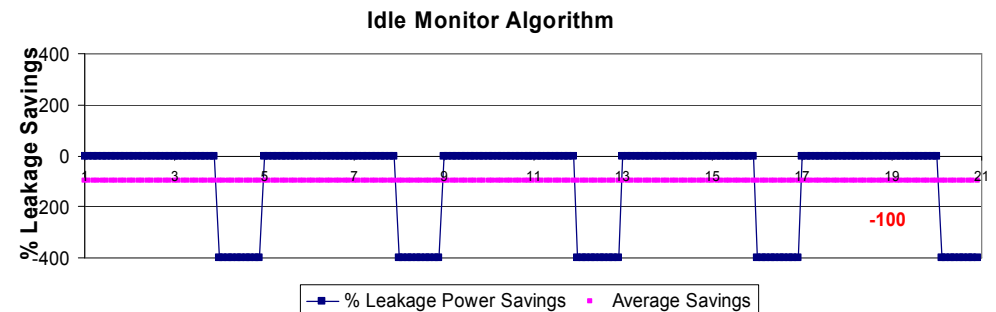
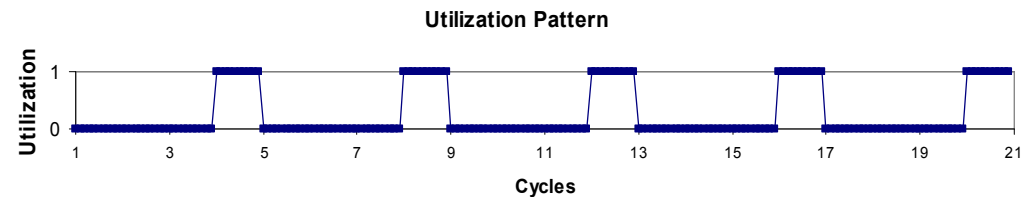
24 22 20 18 16 14 12 10 8 6 4 2 0

Break-even point

Large Potential for Power Gating!

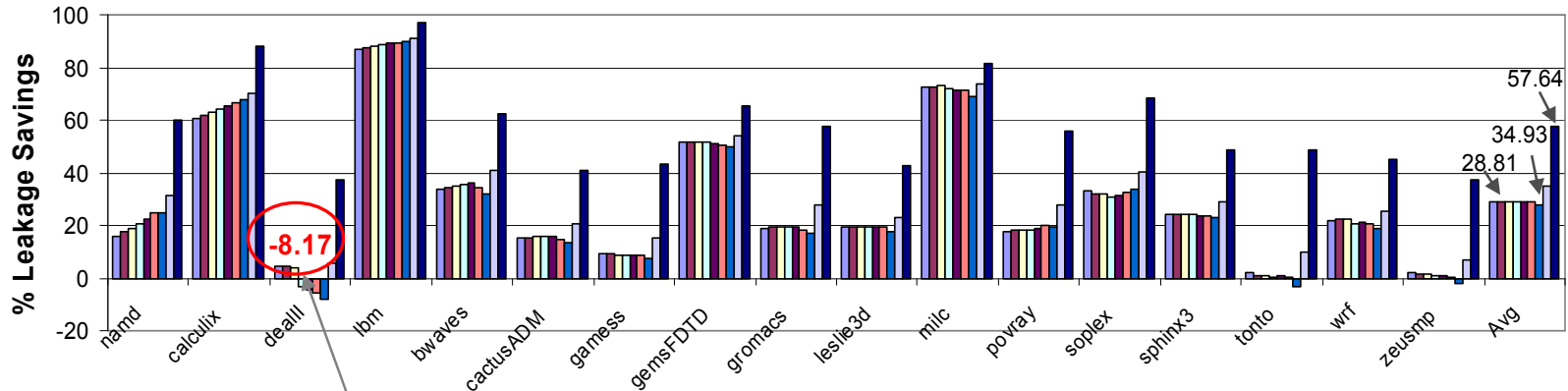
Pitfalls of Current Power Gating Algorithms

- Idle interval prediction can be consistently wrong:
 - => power gating algorithm consistently wastes power instead of saving
- Possible scenarios in loops
 - Idle monitor failure
 - Idle detect 3, break-even 20
 - Average leakage power loss **100%**
 - Utilization monitor failure
 - Utilization threshold 30%
 - Average leakage power loss **98.5%**



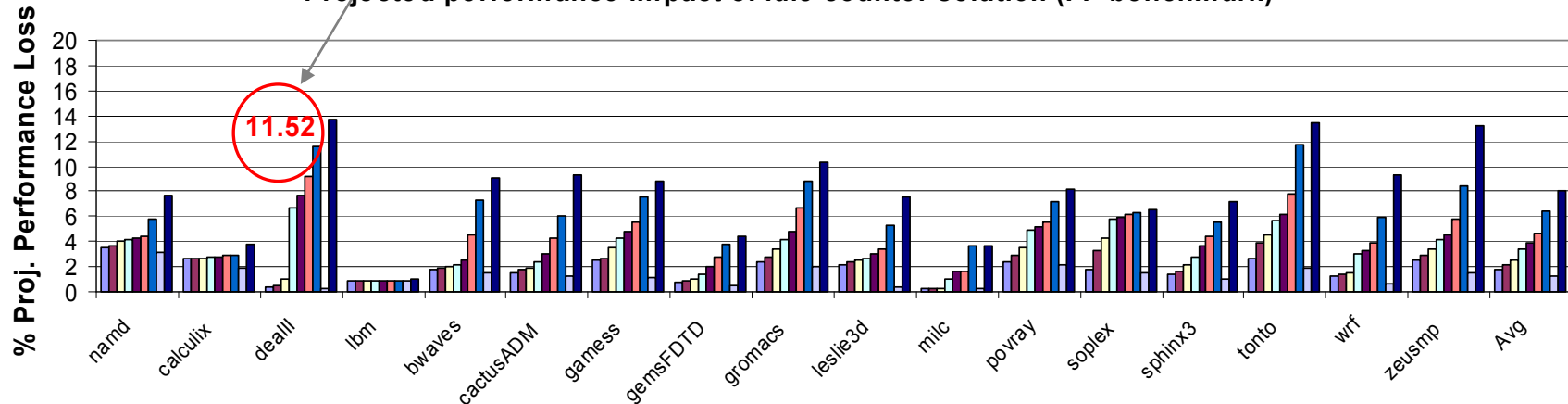
Single Level Idle Detect Power Gating Algorithm

Power savings of idle counter solution function of idle_detect for **FXU0** unit (FP benchmark)



Inefficient Behavior

Projected performance impact of idle counter solution (FP benchmark)



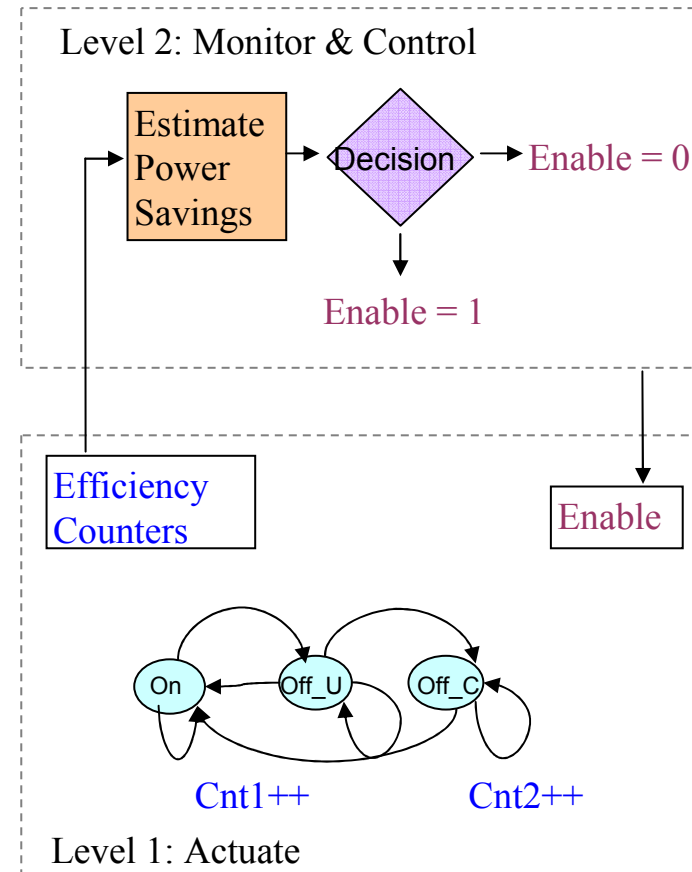
A. Lungu et al., ISLPED-2009

15 13 11 9 7 5 3 Oracle Cycle by Cycle

Idle Detect

Two Level (Guarded) Power Gating Algorithms

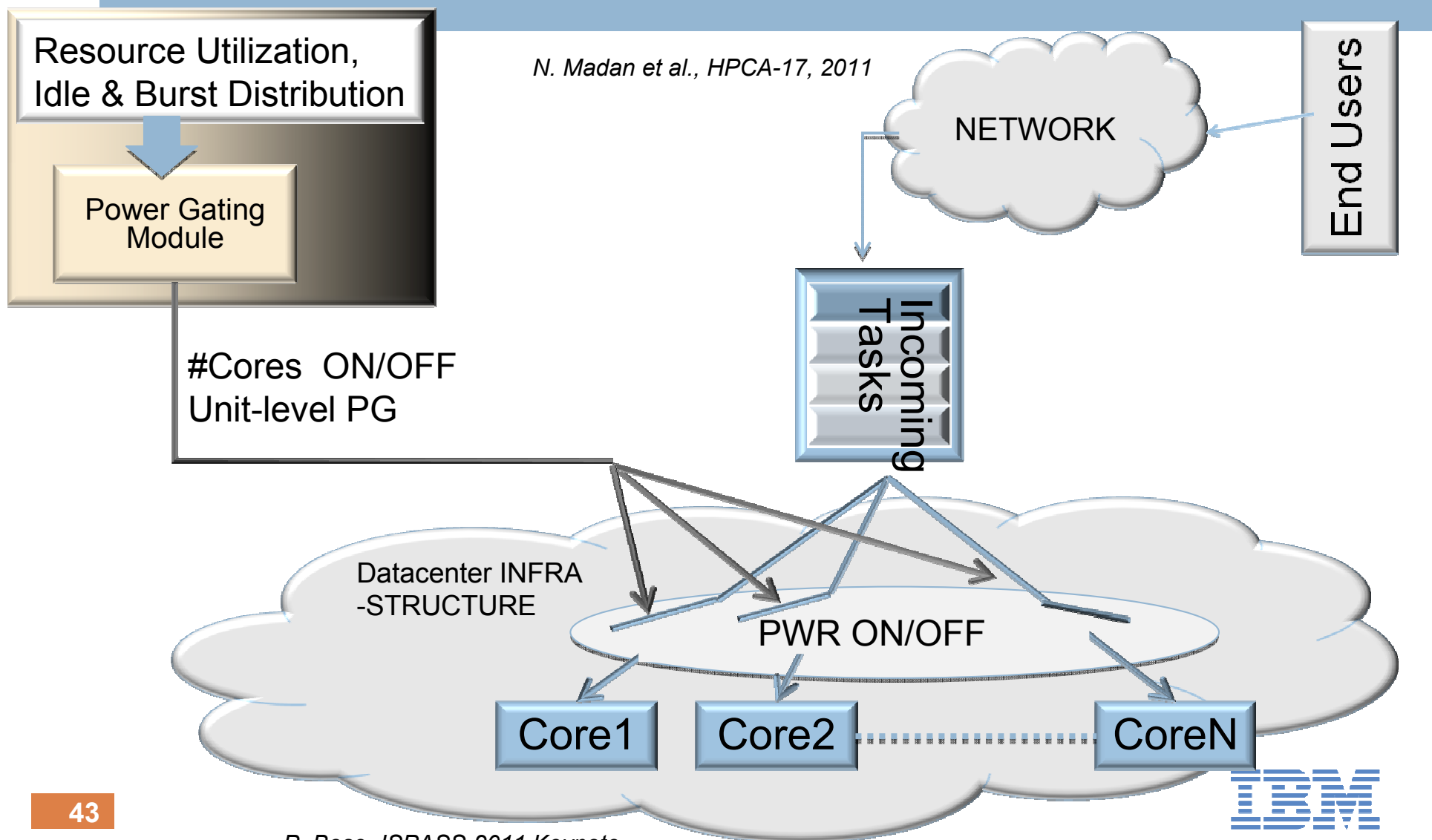
- Observations:
 - Efficiency requirement of power saving schemes: save power
 - Single level idle prediction algorithms can behave incorrectly and waste power
- Target:
 - Improve quality of power gating schemes by reducing or eliminating their risk of wasting power
- Idea:
 - Add second level monitor to control enabling of power gating scheme
 - Improve efficiency of power wasting cases without degrading power saving of the common case



Off_U: Power gated, uncompensated

Off_C: Power gated, compensated

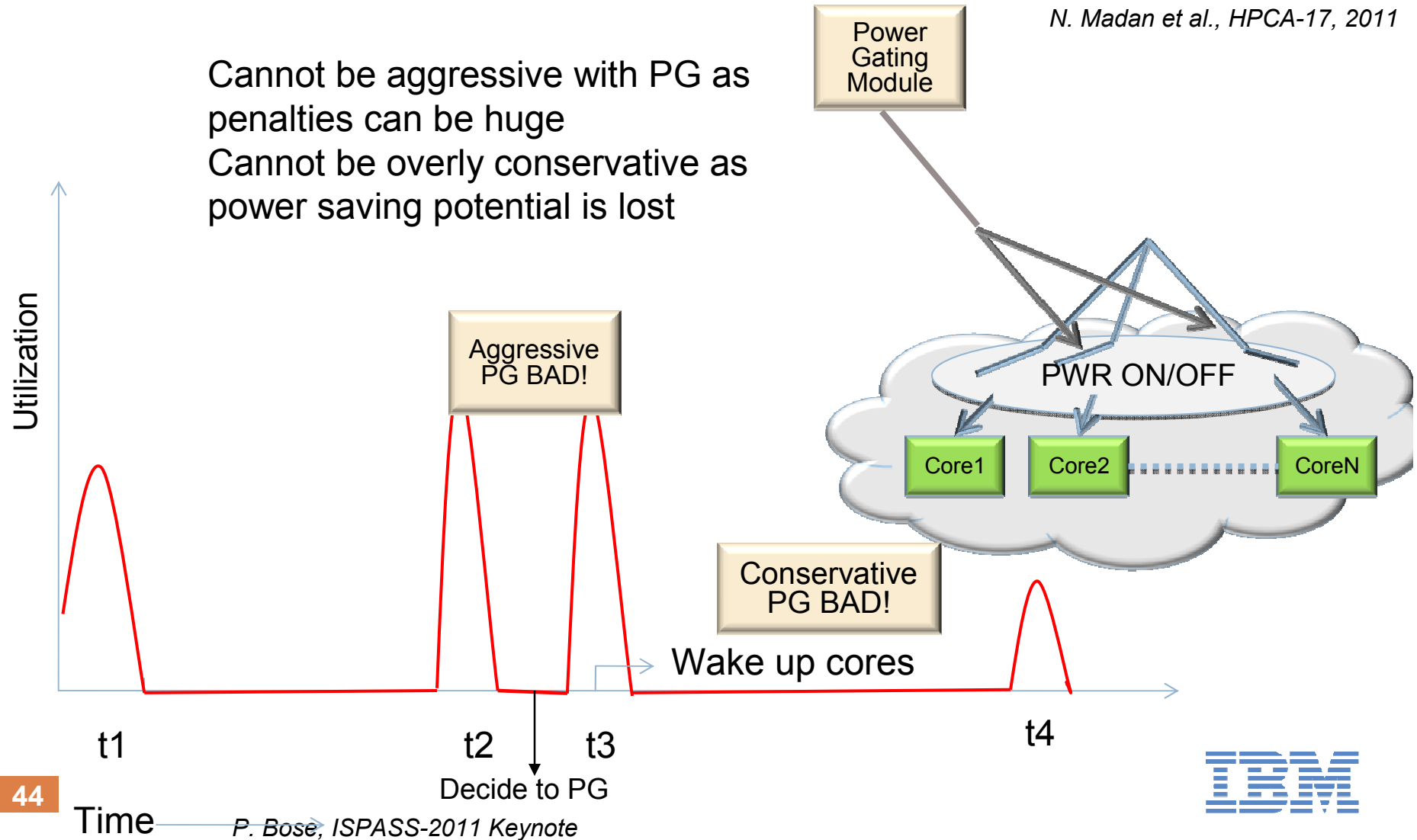
Power Gating in a Datacenter Setting



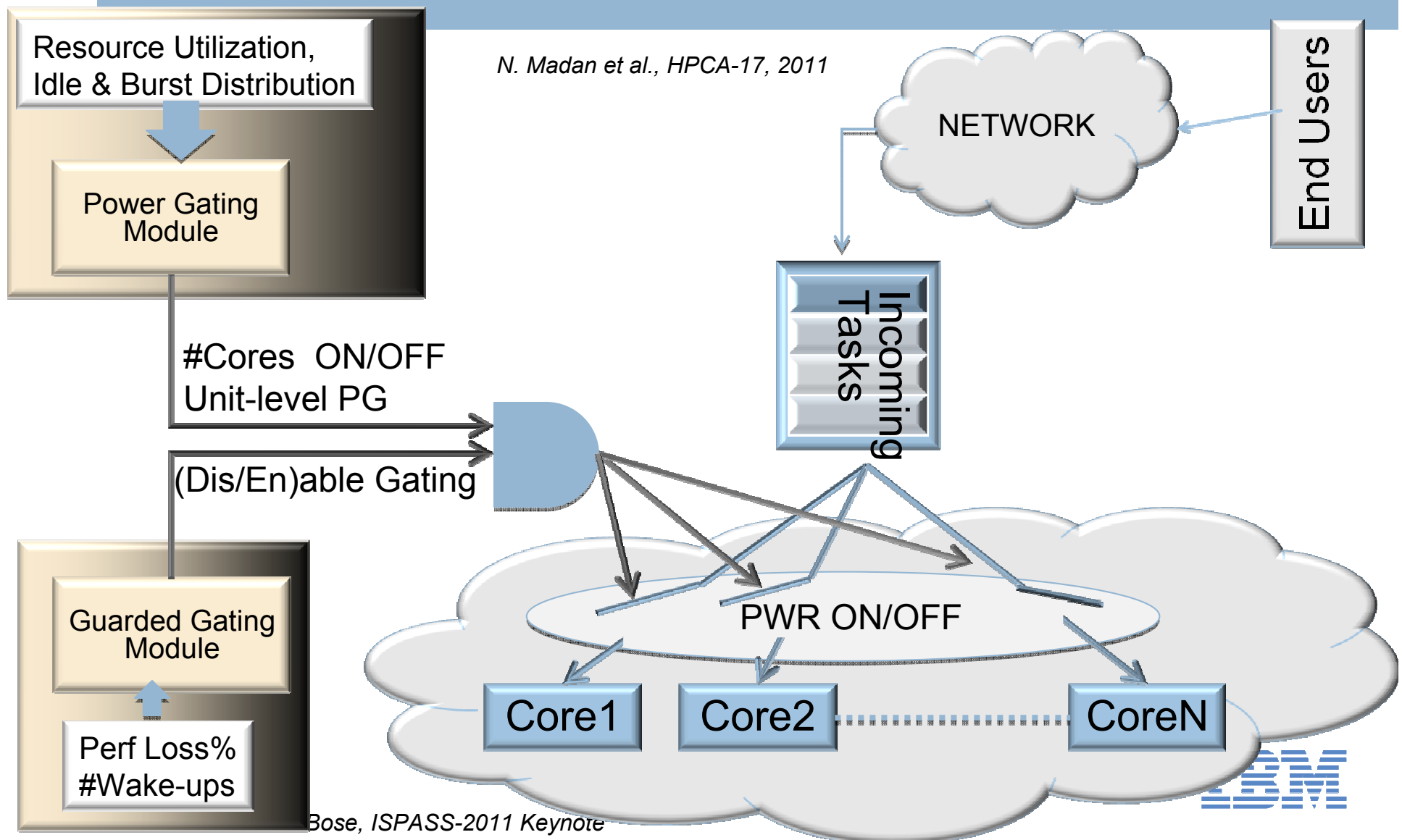
Problems with Core-Level Power Gating

N. Madan et al., HPCA-17, 2011

Cannot be aggressive with PG as penalties can be huge
Cannot be overly conservative as power saving potential is lost

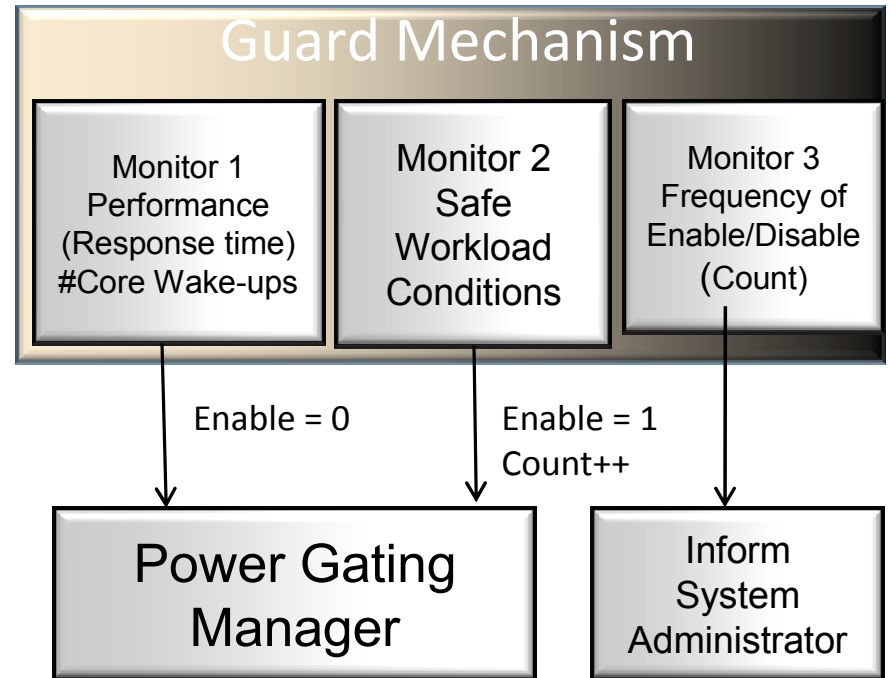


Augmenting Core-Level Power Gating with Guarding



Proposed Guard Mechanism

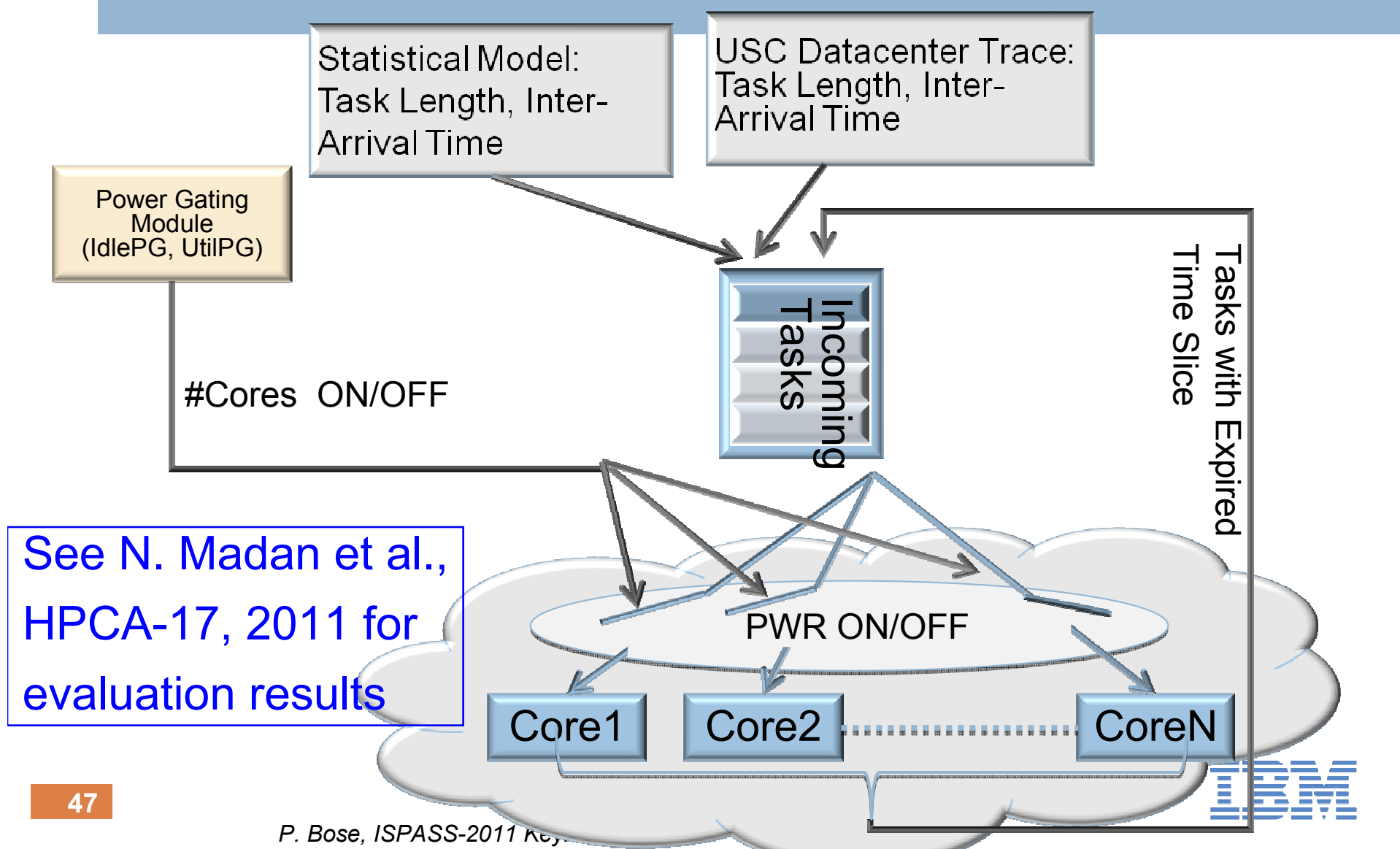
- Monitor system response time
 - ▣ Response time can be very high when the system is overly utilized
- Monitor number of core wake-ups
 - ▣ Wake-up latency and switching power can be negligible too
- Only If **both** monitors show unacceptable behavior
 - ▣ Disable power manager
- Re-enable power manager after a programmable time period
- Alert the system manager



See N. Madan et al., HPCA-17, 2011 for Evaluation Results

More coverage at: [Energy-Secure Architectures: Tutorial at ISCA-2011](#)

Queuing Model Based Evaluation Framework



Concluding Remarks

- **Power and Reliability Walls are Key Impediments to Realization of Extreme Scale Computing Targets of the Future**
 - Reliability may well be the more fundamental obstacle beyond a certain size of the system
- **Integrated Models (power/temperature, performance, reliability) are a Grand Challenge**
 - Analytical abstraction methods are essential for speed
 - Yet, accuracy requirements at core/chip and other component level are more stringent than ever because of the implications of the huge scale (system size)

